

Legal and Regulatory Issues in Life Sciences

Detailed Notes

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Two appendices follow: **Appendix A** lists corrections to the course slides and lecture material, and **Appendix B** is a revision sheet of numbers, traps and likely questions.

PART 1 INTERNATIONAL ETHICAL CODES

1.1 Why the codes exist: the dark history

Before the Second World War there was almost no law on human experimentation. Researchers experimented on themselves or on their own patients, and military research used **prisoners without any concern for consent or wellbeing**. The Nazi camp experiments were the worst: prisoners were placed in compression chambers and freezing water, given gunshot wounds, and subjected to graft transplants between twins to observe the body's response. **Death was the intended endpoint of most of these experiments, and where the subject survived, doctors performed dissection before death to study bodily changes.**

Two ideas emerged and now underpin every code. First, research on human beings must have **social value**, not merely scientific interest. Second, that value never dispenses with the participant's consent.

The modern formulation of the first idea is Guideline 1 of the CIOMS guidelines: *the ethical justification for undertaking health related research involving humans is its scientific and social value, being the prospect of generating the knowledge and the means necessary to protect and promote people's health*. Social value is hard to quantify but rests on three factors.

1. The **quality of the information** to be produced.
2. Its **relevance to significant health problems**.
3. Its **contribution to creating or evaluating interventions, policies or practices** that promote individual or public health.

A working definition to open any answer: **ethics** is the systematic study of values so as to decide what is right and wrong, and **ethics in clinical research** focuses largely on identifying and implementing **the acceptable conditions for exposure of some individuals to risks and burdens for the benefit of society at large**. That last sentence contains the whole moral problem: some people bear the risk, everyone takes the benefit.

1.2 The six codes, and why their legal force differs

1. **Nuremberg Code**, 1947.

2. **Declaration of Helsinki**, 1964, revised eight times, the current version being October 2024.
3. **Belmont Report**, 1979.
4. **United States Common Rule**, 1991.
5. **CIOMS International Ethical Guidelines**, 1982, revised 1993, 2002 and 2016.
6. **India: ICMR Ethical Guidelines for Biomedical Research on Human Subjects**, 2000, revised 2006, now the National Ethical Guidelines of 2017.

Note the difference in legal character, because an examiner will reward it. Nuremberg was a **judgment**. Helsinki is **professional ethics** produced by a body of doctors, not a State. Belmont is a **government commission report**. The Common Rule is **binding regulation**. CIOMS is **international guidance aimed at cross border research**. The Indian guidelines are **administrative guidance that binding rules then adopt by reference**. **Their content overlaps almost completely; their enforceability does not.**

1.3 The Nuremberg Code, 1947

United States v. Karl Brandt and others, Nuremberg Military Tribunal, verdict 20 August 1947

Facts. Twenty three accused, mostly physicians, were tried for experiments conducted on concentration camp inmates. Karl Brandt was the senior medical figure.

Held. The Code was **drafted on 9 August 1947**, and on **20 August 1947** the judges delivered the verdict against Brandt and twenty two others. Expert medical advisers to the prosecution had submitted a memorandum of **six points, which the Tribunal revised to ten**. The ten points appear in the section of the verdict headed *Permissible Medical Experiments*.

Principle. **Human experimentation is justified only if its results benefit society and it is carried out in accordance with basic principles that satisfy moral, ethical and legal concepts.** The Code is said to rest largely on the **Hippocratic Oath**, read as endorsing the experimental approach to medicine while protecting the patient. Source: *Permissible Medical Experiments*, Trials of War Criminals before the Nuremberg Military Tribunals under Control Council Law No. 10, volume 2, pages 181 to 182.

Four organising ideas run through the Code: **informed consent, absence of coercion, properly formulated scientific experimentation, and beneficence towards participants**. The ten points themselves are these.

1. **The voluntary consent of the human subject is absolutely essential.**
2. The experiment should yield **fruitful results for the good of society**, unprocurable by other methods or means of study, and not be random and unnecessary in nature.
3. It should be designed and based on the **results of animal experimentation** and on knowledge of the natural history of the disease, so that the anticipated results justify performing it.
4. It should be conducted so as to **avoid all unnecessary physical and mental suffering and injury**.
5. **No experiment** should be conducted where there is an a priori reason to believe that death or disabling injury will occur, except perhaps where the experimental physicians also serve as subjects.
6. The **degree of risk** should never exceed that determined by the humanitarian importance of the problem to be solved.
7. **Proper preparations and adequate facilities** must protect the subject against even remote possibilities of injury, disability or death.
8. The experiment should be conducted **only by scientifically qualified persons**, with the highest degree of skill and care at every stage.
9. The subject must be **at liberty to bring the experiment to an end** if he has reached the physical or mental state where continuation seems impossible.
10. The **scientist in charge must be prepared to terminate** the experiment at any stage if, in the exercise of good faith, superior skill and careful judgement, he has probable cause to believe that continuation is likely to result in injury, disability or death.

Point 3 is the bridge to Part 2. **Nuremberg made prior animal experimentation an ethical precondition of human research, which is why animal ethics belongs to this subject and not to a separate one.**

One distinction is asked repeatedly. Nuremberg required **voluntary** consent, meaning consent free of force, fraud, duress or pressure. Modern law requires **informed** consent, adding that the participant must be told the purpose, the procedures, the risks, the expected benefits, the alternatives, the compensation arrangements, the right to withdraw, and even any intellectual property arising from the research.

1.4 Declaration of Helsinki, 1964

The **Willowbrook State School study** on the natural course of infective hepatitis in children, and the **Jewish Chronic Disease Hospital study** on the body's ability to reject cancer cells in debilitated subjects, showed that the Nuremberg Code had not stopped unethical research. The **World Medical Association** responded with guidelines to safeguard the rights and wellbeing of participants, adopted by the **18th WMA General Assembly in 1964**.

The version described in the course material is the **2000 Edinburgh revision, adopted at the 52nd WMA General Assembly**, containing **32 principles**, which stress five things.

1. **Informed consent.**
2. **Confidentiality of data.**
3. Protection of **vulnerable populations**.
4. The requirement of a **written protocol**.
5. **Review of the protocol, including the scientific reasons for the study, by an ethics committee.**

Correction. The Declaration has been revised eight times and the current version is the revision adopted at Helsinki on 19 October 2024 at the 75th WMA General Assembly. Every earlier version, including the 2000 text, is superseded except for historical purposes. The 2024 revision treats vulnerability as **situational and dynamic** rather than as a fixed list of groups, requires **community engagement**, adds the pursuit of **global justice** and of **scientific rigour and integrity**, and calls the subject a **participant**. Present the 2000 text as the classic formulation and then state the 2024 position.

Helsinki's most durable institutional contribution is the ethics committee. **The need for an ethics committee first appeared in the Declaration of Helsinki in 1964; in India it first appeared in the ICMR Policy Statement of 1980.**

1.5 Belmont Report, 1979

The trigger was the **Tuskegee Syphilis Study**, begun in **1932 with 399 syphilitic African American men** to observe the natural course of syphilis. It was meant to last about six months, but because the researchers were getting good data they continued it. **Participants were misled and deprived of treatment even after penicillin became available in the 1940s.** The study was exposed in **1972**, by which time it had caused **28 deaths, permanent disability in 100 subjects, infection of 40 wives and 19 cases of congenital syphilis.**

The **National Commission for the Protection of Human Subjects of Biomedical and Behavioural Research** was formed to investigate, and wrote the **Belmont Report in 1979**, which laid the foundation for United States regulation of human subjects research. Belmont states **three** principles, each with a matching application, and that pairing is what examiners test.

1. **Respect for persons**, applied as **informed consent**.
2. **Beneficence**, applied as **assessment of risks and benefits by ethics committees**.
3. **Justice**, applied as **selection of subjects**.

1.6 CIOMS

The **Council for International Organizations of Medical Sciences** is an international, non governmental, non profit organisation **established jointly by the World Health Organization and UNESCO in 1949**. It represents a substantial part of the biomedical community through member organisations including national academies of sciences and medical research councils, and its mission is to advance public health through guidance on health research, including ethics, medical product development and safety.

In **1982**, with the World Health Organization, it produced the **International Ethical Guidelines for Biomedical Research Involving Human Subjects**, revised in 1993, 2002 and 2016. **Its distinctive contribution is that it addresses research funded by rich countries and conducted in poor ones, which makes it the most directly relevant international instrument for India.** Five concerns recur.

1. The **investigator's duties regarding consent** where literacy and health awareness are low.
2. **Appropriate inducements**, meaning that payment must not be so large as to destroy voluntariness in a poor community.
3. **Special and vulnerable populations.**
4. **Therapeutic misconception**, where the participant believes the trial is treatment.
5. **Post trial access** to a drug shown to be beneficial.

1.7 United States Common Rule, 1991

The Common Rule began as a **1981 rule of ethics** on biomedical and behavioural research involving human subjects, governing Institutional Review Boards and following the 1975 Tokyo revision of Helsinki. It was consolidated when the Department of Health and Human Services published the **Federal Policy for the Protection of Human Subjects in 1991**. Four main elements.

1. Requirements for **assuring compliance by research institutions.**
2. Requirements for researchers **obtaining, waiving and documenting informed consent.**
3. Requirements for **Institutional Review Board membership, function, operations, review of research and record keeping.**
4. **Additional protections for vulnerable subjects:** Subpart B for pregnant women, in vitro fertilisation and fetuses; Subpart C for prisoners; Subpart D for children.

1.8 The seven principles of ethical research

Alongside the codes, seven principles are said to guide ethical research generally. They compress every code into one checklist and make a clean opening paragraph for an essay.

1. **Social and clinical value.**
2. **Scientific validity.**
3. **Fair subject selection.**
4. **Favourable risk and benefit ratio.**
5. **Independent review.**
6. **Informed consent.**
7. **Respect for potential and enrolled subjects.**

1.9 India: the ICMR line and the Indian controversy

In **February 1980** the Indian Council of Medical Research released a **Policy Statement on Ethical Considerations involved in Research on Human Subjects**. **This was the first policy statement giving official guidelines for the establishment of ethics committees in all medical colleges and research centres in India.**

The Indian example matters because it is home grown rather than borrowed. **In the 1970s and 1980s researchers at the Institute for Cytology and Preventive Oncology, New Delhi, carried out a study on 1,158 women patients at different stages of cervical dysplasia, that is precancerous lesions of the cervix.** The patients were **left untreated** so that the researchers could observe how many lesions progressed to cancer and how many regressed. By the end **71 women had developed malignancies**, in **nine** the lesions had progressed to **invasive cancer**, and **62 women were treated only after developing localised cancer.**

The controversy became public in **1997**. The Council then developed the **Ethical Guidelines for Biomedical Research on Human Subjects**, finalised in **2000** and revised in **2006**, which elaborate Belmont's three principles by **inducting twelve general principles**.

The twelve principles: the short list

1. **Essentiality.**
2. **Professional competence.**
3. **Voluntariness.**
4. **Maximisation of benefit.**
5. **Non exploitation.**
6. **Institutional arrangements.**
7. **Social responsibility.**
8. **Transparency and accountability.**
9. **Ensuring privacy and confidentiality.**
10. **Totality of responsibility.**
11. **Risk minimisation.**
12. **Environmental protection.**

The twelve principles: the elaborated text

1. **Essentiality.** The research must be essential for the advancement of knowledge benefiting patients, doctors and others in health care, and also for the **ecological and environmental wellbeing of the planet**.
2. **Voluntariness, informed consent and community agreement.** The participant must know the nature of the research and its probable consequences and must choose **without the influence of the treating doctor**. Where a community is the participant, these principles apply both to the community as a whole and to each member.
3. **Non exploitation.** Participants must be remunerated and made aware of all risks irrespective of social, economic or educational condition. Each protocol must provide **compensation through insurance or otherwise for all foreseeable and hidden risks**.
4. **Privacy and confidentiality.** Data must be kept confidential and not disclosed without valid legal or scientific reason.
5. **Precaution and risk minimisation.** Due care at every stage from the research idea to subsequent applicative use. **The ethics committee has an active role in risk minimisation.**
6. **Professional competence.** Research only by competent and qualified persons in their fields.
7. **Accountability and transparency.** Fair, honest, impartial and transparent conduct after full disclosure of interests, with **research data retained for a minimum of five years** for scrutiny by the appropriate authority.
8. **Maximisation of public interest and distributive justice.** Results used for the benefit of all, especially the participants and their community, and not only the socially better off.
9. **Institutional arrangements.** All institutional arrangements made transparently.
10. **Public domain.** Results published, and **detailed information about clinical trials made public before recruitment begins** through a registry offering free online access, such as the Clinical Trials Registry of India.
11. **Totality of responsibility.** Professional and moral responsibility on all directly or indirectly connected with the research.
12. **Compliance.** Compliance with the guidelines specific to the area of research.

The current version is the **National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017**, which rule 3 of the New Drugs and Clinical Trials Rules, 2019 requires every researcher to follow. **The guidelines therefore now carry indirect statutory force.**

1.10 Informed consent in practice

The operative documents are a **Participant or Patient Information Sheet** attached to an **Informed Consent Form**, and **both must be approved by the ethics committee before use**. The information sheet must contain the following.

1. A statement that the study **involves research**.
2. The **purpose** of the research and the **expected duration** of participation.
3. A description of the **procedures**, identifying any that are **experimental**.
4. Any **reasonably foreseeable risks or discomforts**.
5. Any **benefits** to the subject or to others.
6. The **trial treatment schedule** and the **probability of random assignment** to each treatment, especially in randomised placebo controlled trials.
7. Disclosure of appropriate **alternative procedures or courses of treatment** that might be advantageous.
8. The extent to which **confidentiality of records** will be maintained.
9. For research involving more than minimal risk, whether **compensation** and **medical treatment** are available if injury occurs, what they consist of, and where further information may be obtained.
10. **Whom to contact** about the research, about participants' rights, and in the event of a research related injury.
11. A statement that **participation is voluntary**, that refusal involves no penalty or loss of benefits, and that the subject may **discontinue at any time** without penalty.

1.11 The ethics committee

Also called the Institutional Review Board or Ethics Review Board, it **stands as the bridge between the researcher and the ethical guidelines of the country**. Four structural points.

1. Establishment requires **five to fifteen members**.
2. Membership must include at least **one basic medical scientist, preferably a pharmacologist; one clinician; a legal expert; a social scientist or representative of a non governmental organisation or a philosopher or theologian; and a lay person from the community**.
3. **Every institute where research is going on should have its own committee**.
4. **The head should preferably be from outside the institute**, which is the structural guarantee of independence.

Under the New Drugs and Clinical Trials Rules, 2019 the committee must also be **registered**, and an unregistered committee cannot validly approve a trial. It reviews serious adverse events, which must be reported within **fourteen days**, and the compensation payable, which is computed under the **Seventh Schedule**, with payment in a trial related death due within **thirty days** and **free medical management** for as long as required.

PART 2 ETHICS IN ANIMAL RESEARCH

2.1 Purpose and the two threshold questions

Animals are used for three purposes: **development of medicines and vaccines, fundamental biological research**, and **toxicology and safety studies**. Before any of it, two questions must be asked and answered on the record.

1. **Must this project involve animal studies at all?**
2. **Must this particular test be carried out?**

2.2 Why animal data is legally required, not merely customary

Animal experimentation is a step in getting a drug approved for human use, and animal data enters the regulatory process at four distinct points.

1. **Basic life science and behavioural research** relies on animal models.
2. **Preclinical testing** of drugs precedes first administration to humans.
3. **New drug applications must include animal data.**
4. Animal data is sometimes included **when eliciting informed consent** from human participants, because the risk cannot otherwise be described.

So the law simultaneously requires animal use and requires it to be minimised. That tension is the entire subject.

2.3 The ethical dilemma, in four propositions

1. It is **morally imperative** to find treatments for new or serious human ailments.
2. **Live animal research is at present the only way** to bring some effective medicines.
3. **Live animal research will always cause animals to suffer.**
4. The suffering **does not benefit the animals**; the benefit accrues to the human race and should in some form be compensated.

2.4 Three ethical theories

Contractarianism

From the Hobbesian line of social contract thought. Persons are primarily self interested, and a rational assessment of the best strategy for maximising self interest leads them to act morally, where moral norms are determined by the maximisation of joint interest, and to consent to governmental authority. **The theory applies only to individuals who can contract into the moral community, and since animals cannot, animal experimentation is not an ethical issue on this view at all.** Use it to explain historical practice, not to justify it.

Utilitarianism

An **interest** is a capacity to reduce suffering or enhance enjoyment. Individuals, including animals, have an interest in acts that enhance enjoyment or reduce suffering. The theory always favours bringing something beneficial to society, so **animal research is ethically allowed**. But because animal suffering counts in the calculation, the theory creates a positive duty to reduce it, and utilitarians act towards the benefit of animals by three means: **replacement of live animals, reduction in the number of samples, and refinement of techniques.** **The 3Rs are therefore utilitarianism converted into practice, and then into law.** Peter Singer is the modern figure associated with this line.

The rights view

A sentient being cannot be treated merely as a means to achieve another's goals. Associated with Tom Regan, the position is categorically abolitionist: laboratory animals are not our tasters and we are not their kings, and because they are treated as though their value were reducible to their usefulness, their rights are violated as much in promising research as in trivial research. Bernard Rollin and Arthur Caplan complete the group of four contemporary ethicists named in this course.

Two quotations carry the argument in either direction. Jeremy Bentham asked not *can they reason* nor *can they talk* but ****can they suffer****, which is the origin of the law's focus on sentience rather than rationality. Immanuel Kant held that man can have no duties to beings other than man, because the obligating subject must be a person; he still condemned cruelty, but as a wrong that degrades the human being rather than as a wrong to the animal.

2.5 Three questions the examiner will ask

1. In what way does the use of animals in research represent a direct conflict between the protection of human beings and of animals?
2. Even if we believe we should never intentionally inflict pain or suffering on animals, what if such treatment would lower the risk of inflicting pain or suffering on humans?
3. If we have to choose between an animal and a human being as the subject of a research project, is it ever justified to use a human subject instead of an animal?

Answer the first by distinguishing interests from rights, the second through the utilitarian calculus plus the 3Rs, and the third by pointing out that the law has already answered it: Nuremberg point 3 requires the animal study first, so the human subject is used only afterwards.

2.6 Two controversial acute toxicity tests

1. **The LD50 test, meaning lethal dose fifty per cent.** It evaluates toxicity by determining the dose required to kill fifty per cent of the test animal population.
2. **The Draize test.** A test substance is applied to an animal's **eyes or skin, usually an albino rabbit**, to assess irritancy.

Two set questions follow. In what way do these tests represent a direct conflict between the rights and interests of humans and animals? **And would it be justified to ban these tests, even if that would increase the risk of humans being hurt or killed by exposure to toxic substances?** Resolve it through validated alternatives and the 3Rs rather than by choosing a side, and remember that LD50 data is used for household cleaners, phenyl and pesticides as much as for medicines, and is what makes poisoning treatment protocols possible.

2.7 The 3Rs

1. **Replacement** of live animals, by cell and tissue culture, computer simulation, organ on chip systems and human volunteer studies where ethical.
2. **Reduction** in the number of samples to the statistical minimum, which is why a committee demands a power calculation.
3. **Refinement** of techniques, through better anaesthesia and analgesia, less invasive procedures, humane endpoints and better housing and handling.

2.8 United States framework

The **Animal Welfare Act** is described in the course material as the Act of 1985 and as **the first federal law in the United States regulating animals in research**. For accuracy, the Act was enacted in **1966**, and 1985 is the year of the amendment that introduced the requirements below. Write that it is the Animal Welfare Act as amended in 1985. Its six requirements are these.

1. **Balancing societal benefits** against the imposition on animals, in terms of suffering and of numbers used.
2. **Minimisation of pain, suffering and distress** in all procedures.
3. **Species appropriate housing, feeding and care.**
4. **Involvement of veterinary personnel.**
5. **Qualifications of investigators** and their personnel.
6. **Oversight by an Institutional Animal Care and Use Committee.**

The Institutional Animal Care and Use Committee

The committee must **evaluate, report on and inspect the facilities** of all units employing animals for research, teaching or testing. **Its mission is to assure the humane handling, care, treatment and transportation of animals.** Three criteria govern its evaluation.

1. Provide a **clear rationale** for animal use.
2. **Justify the number** of animals used in the study.
3. Conduct a **risk and benefit analysis**.

A sharp critique is available: the official United States count of about 1.1 million animals used in 2009 **excludes rats and mice**, which are the overwhelming majority of laboratory animals, because those species fall outside the Act. The published figure therefore understates use by an order of magnitude.

2.9 Indian framework

The regulator is the **Committee for the Purpose of Control and Supervision of Experiments on Animals**, a body **formed under the Prevention of Cruelty to Animals Act, 1960** and constituted under **section 15**, with its duties and powers in **section 17**. The course material records that it was **formed in 1964 and revived in 1998**.

Its objective is to ensure that animals are not subjected to unnecessary pain or suffering before, during or after experiments on them. Memorise that phrase; it appears verbatim in official documents.

The operative rules are the **Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998**, amended in **2001 and 2005**, the second amendment notified in January 2006, which is why both years appear in the literature.

Two current corrections. **The body has been renamed the Committee for Control and Supervision of Experiments on Animals, and animal welfare was transferred from the Ministry of Environment, Forest and Climate Change to the Department of Animal Husbandry and Dairying in April 2019.**

Compliance has four limbs. An establishment doing biomedical research, or breeding or trading laboratory animals, must **register** with the central committee, must constitute an **Institutional Animal Ethics Committee** including a nominee of the central committee, must have its **animal house inspected**, and must obtain **approval for every project**. **Approval given by an improperly constituted institutional committee is invalid.**

2.10 The 2023 shift towards non animal methods

A current development that completes this topic and that the lecture slides predate. In **March 2023** the Government amended the New Drugs and Clinical Trials Rules, 2019 by the **New Drugs and Clinical Trials (Amendment) Rules, 2023**, made after public comments and in consultation with the Drugs Technical Advisory Board. **The amendment authorises researchers to use non animal and human relevant methods to test the safety and efficacy of new drugs.** The named alternatives are **three dimensional organoids, three dimensional bioprinting, organs on chip systems, and advanced computational methods**.

Place it in its international context: the European Union adopted a resolution on an action plan for the transition away from animal use in 2021, the United States passed the **FDA Modernization Act 2.0 in December 2022**, and Canada amended its environmental protection legislation in June 2023 to replace, reduce or refine the use of vertebrate animals in toxicity testing.

The realistic assessment, and the one to give in an answer: **the permission exists but the guidance does not.** Reporting through 2026 indicates that the absence of detailed regulatory guidance on acceptance criteria and validation pathways has limited routine use of these methods in formal approval submissions, and that Indian manufacturers still face animal toxicity requirements in categories where the United States and the European Union permit waivers. Legally, this is a classic case of an enabling amendment without implementing standards.

2.11 Indian case law on animals

Animal Welfare Board of India v. A. Nagaraja, (2014) 7 SCC 547 (Supreme Court, 7 May 2014)

Facts. Jallikattu in Tamil Nadu and bullock cart races in Maharashtra were challenged, together with the Tamil Nadu Regulation of Jallikattu Act, 2009, following a notification of 11 July 2011 barring the use of bulls as performing animals.

Held. The events as then conducted violated **sections 3 and 11** of the Prevention of Cruelty to Animals Act, 1960. The notification was upheld and the State Act was held to offend the central Act.

Principle. The 1960 Act must be read with **Articles 51A(g), 48A and 21**. The Court adopted the **five internationally recognised freedoms** and held that **animals have a right to live with dignity**.

Animal Welfare Board of India v. Union of India, 2023 SCC OnLine SC 661 (Supreme Court, Constitution Bench, 18 May 2023)

Facts. Tamil Nadu, Maharashtra and Karnataka amended the 1960 Act to permit Jallikattu, bullock cart races and Kambala with safeguards. The amendments were challenged.

Held. The amendments were **upheld**. They substantially reduce pain, fall within **Entry 17 of the Concurrent List**, and cure the defects identified earlier. The Article 29(1) cultural rights question was left open.

Principle. **Read together, the two cases show that Indian law protects animal welfare rather than conferring animal rights**, and that a legislature may lawfully restore a practice by curing the defects a court has identified. *Nagaraja* was confined and distinguished, not overruled.

PART 3 BIOETHICS

3.1 Definitions

1. **Ethics** is the systematic study of values, so as to decide what is right and what is wrong.
2. **Bioethics** is a branch of applied ethics that studies the **philosophical, social and legal issues arising in medicine and the life sciences**.
3. **Clinical ethics**, or ethics in clinical research, focuses on the acceptable conditions for exposing some individuals to risks and burdens for the benefit of society at large.

The standard analytical framework, associated with Beauchamp and Childress and adopted by the ICMR in its Code of Conduct for Research Scientists, has **four principles**: **autonomy**, meaning respect for persons including consent, privacy and confidentiality; **beneficence**, meaning fruitful results and doing good; **nonmaleficence**, meaning no deliberate harm; and **justice**, meaning equitable distribution of risks and benefits. **Do not confuse this four principle framework with Belmont's three principles. Belmont folds nonmaleficence into beneficence.**

3.2 Mapping the field: the colour code of biotechnology

A classification attributed to Kafarski (2012) sorts biotechnology subsectors by colour, the basis of delineation being either the source of the biomaterial or the processes and markets involved.

1. **Blue**, marine, delineated by the **source of biomaterial**.
2. **Red**, medical and pharmaceutical.
3. **Green**, agricultural.
4. **Yellow**, nutritional.
5. **White**, industrial.
6. **Grey**, environmental protection.
7. **Brown**, management of deserts and arid regions.
8. **Gold**, bioinformatics, computer science and chip technology.
9. **Violet**, **law, ethical and philosophical issues**.

10. **Dark**, bioterrorism and biological weapons.

Two entries matter beyond classification. **Violet is this subject.** And **dark** is why dual use is a legal category and not merely a moral worry.

3.3 The five major challenges in health biotechnology

1. **Ethical issues** generally.
2. **Protection of human subjects during clinical trials.**
3. **Privacy.**
4. **Stem cell research.**
5. **Bioterrorism.**

The governing question posed in the lecture is worth quoting: **is it desirable to intervene with the fabric of life, when technological compulsion will drive developments ahead of proper ethical consideration of their propriety?** That states the precautionary problem in bioethics exactly as the precautionary principle states it in environmental law.

The commercial backdrop explains why the pressure exists. Research and development takes **seven to ten years** for a biotechnology product, and the sector faces **loss of patents bringing competition from biosimilars and generics, high development costs, low research productivity, and a strong need for mergers and acquisitions to strengthen the pipeline.** Public support varies sharply between medicinal, industrial and agricultural biotechnology.

3.4 The specific problems

Dual use

Dual use biotechnology is technology where both good and ill use are possible. Three current examples are named: **gene editing, synthetic biology, and CRISPR technology.** India's legal expression of this concern is the ICMR Code of Conduct for Research Scientists, whose first and third principles require that no research assist bioterrorism or biowarfare and that dual use information be restricted to those who need to know. The Code operates by a **written undertaking signed by each scientist**, not by penalty, and its aim is that all research involving microbial or other biological agents or toxins is **only of types and in quantities that have justification for prophylactic, protective or other peaceful purposes**, a deliberate echo of the Biological and Toxin Weapons Convention.

The eleven principles of the ICMR Code of Conduct

1. **Nonmaleficence**, so that no discovery harms humans, animals, plants or the environment and no one assists bioterrorism or biowarfare.
2. **Beneficence**, so that legitimate benefits outweigh risks and harms.
3. **Risk minimisation**, restricting dual use information to those who need to know and reporting misuse risks to the authorities.
4. **Confidentiality**, with disclosure only to legitimate persons under a contract prescribed by law.
5. **Ethical review** of all relevant research for ethics and safety.
6. **Transmission of ethical values** to everyone who joins the research.
7. **Voluntariness**, which here means **the researcher's right to abstain from work he finds ethically objectionable, not the participant's consent.**
8. **Compliance** with the laws applying to the conduct of science.
9. **Institutional arrangements** assuring biosafety and biosecurity, with transparent and limited access to agents usable as weapons.
10. **Totality of responsibility**, spreading duty to everyone directly or indirectly connected with the research.
11. **Research integrity**, covering reproducibility, honest data collection and analysis, avoidance of fabrication, falsification and plagiarism, proper authorship credit and protection of young scientists.

Pharmacogenomics, gene therapy and designer babies

Pharmacogenomics concerns **individual response to drugs based on genetic make up**. Its legal significance is that finding the right drug and dose for a patient is **currently done on a trial and error basis**, so the technology promises to replace a process that is itself uncontrolled experimentation. It raises genetic privacy and discrimination questions that Indian law does not yet address.

Gene therapy involves genetic engineering of a patient's genetic code to remove or replace a mutant gene causing disease. The stated ethical concern is **potential misuse to create designer babies**, and beneath it lie two harder problems: **the boundary between therapy and enhancement**, and the fact that **interventions in the genome may be passed on to future generations who cannot consent**. **Preimplantation genetic diagnosis** allows viable embryos to be screened for traits such as sex linked diseases before implantation, which in India intersects directly with the law on prenatal sex selection.

Recombinant medicine as the counterpoint

The benefits are equally concrete, and an answer that gives only risks is unbalanced. **Insulin was the first recombinant protein approved, in 1982**, followed by growth hormone and blood clotting Factor VIII. The **Human Genome Project** underlies the whole field.

3.5 Stem cell research: the science in short

Regenerative medicine describes approaches to illness, injury and disability focused on the growth, replacement and repair of cells, organs and tissues specific to the needs of particular individuals. Its branches are stem cells generally, **pluripotent embryonic stem cells**, **pluripotent stem cells produced by genetic reprogramming**, and **multipotent stem cells from fetal, perinatal, paediatric and adult tissue**. Stem cells divide into **somatic or adult cells**, which may be unipotent or multipotent, and **embryonic cells**, which are pluripotent, with **induced pluripotent cells** produced by reprogramming somatic cells.

3.6 The ethical fault line: adult cells against embryonic cells

Adult stem cells occur in many tissues and can differentiate into specialised cells of their tissue of origin and transdifferentiate into cells characteristic of other tissues. Haematopoietic stem cells, for example, can become all three blood cell types and also neural stem cells, cardiomyocytes and liver cells. They can be isolated through plasmapheresis, are already used to treat haematological malignancies and to modify chemotherapy side effects, and can be banked from cord blood for allogeneic and autologous transplantation. **Adult stem cells and cord blood stem cells do not raise special ethical concerns and are widely used in research and clinical care.**

Human embryonic stem cell lines can be derived from the inner cell mass of the five to seven day old blastocyst. Such research is ethically and politically controversial because it involves the destruction of human embryos. Three questions follow: what is the moral status of the embryo, at what stage does human life begin, and is an embryo a person with the same moral status as an adult or a live born child?

The three positions

1. **Human life begins at conception.** On this view, removing the inner cell mass from a blastocyst to derive a cell line is **tantamount to murder**.
2. **The embryo becomes a person in a moral sense at a later stage than fertilisation.** Note the qualification in the lecture: few people believe the embryo or blastocyst is *just* a clump of cells usable without restriction.
3. **The middle ground**, that the early embryo deserves **special respect as a potential human being** but may be used for certain research provided there is good scientific justification, careful oversight, and informed consent from the woman or couple donating it.

The middle ground is the position Indian and most Western regulation has taken, and the fourteen day rule is its concrete legal expression. Note also that opposition is often, but not always, aligned with opposition to

abortion: several prominent pro life figures have supported research on **frozen embryos left over after infertility treatment** that the couple has decided not to donate to another couple.

3.7 Four specific consent problems

Frozen embryos donated for research

Three issues arise: **informed consent from the woman or couple donating**, **consent from the gamete donors** who created the embryo, and **confidentiality of donor information**.

Oocyte donation for research

1. **Medical risks of retrieval**, including ovarian hyperstimulation syndrome, bleeding, infection and complications of anaesthesia, requiring careful evaluation.
2. **Protecting the reproductive interests of women in infertility treatment**, whose reproductive needs must be prioritised over the research.
3. **Payment**. Payment beyond reasonable out of pocket expenses is controversial, being the classic inappropriate inducement.
4. **Quality of informed consent**, where understanding of the process is often poor.

Somatic cell nuclear transfer

Reprogramming somatic cells to produce induced pluripotent cells **avoids the ethical problems specific to embryonic cells**. One route is somatic cell nuclear transfer, the technique that produced Dolly the sheep, in which nuclear DNA from a donor cell is transferred into an oocyte whose nucleus has been removed. Three objections.

1. **Creating embryos specifically for research** is said to violate respect for nascent human life. The counter argument is that pluripotent entities created by this technique are **biologically and ethically distinct from embryos**.
2. **Human reproduction using the technique**. Because of reprogramming errors, cloned animal embryos fail to activate key embryonic genes and newborn clones misexpress hundreds of genes, and error rates in humans would be high. It is also said to violate human dignity and undermine fundamental moral, religious and cultural values, and **a cloned child is the genetic twin of that parent**.
3. **Use of animal oocytes with human DNA**, producing cytoplasmic hybrid embryos, which raises the distinct problem of the hybrid chimera.

Fetal cells and downstream research

Pluripotent cells can be derived from fetal tissue after abortion, which is controversial because of the association with abortion and the risk of **influence on the woman's decision**. Separately, some downstream uses may be so sensitive that the original somatic cell donor might not have agreed: **genetic modification of the cells; injection of derived cells into non human animals, including into their brains; large scale genome sequencing; sharing cell lines with other researchers; and patenting discoveries and developing commercial tests and therapies with no sharing of royalties with donors**. The last is where bioethics meets intellectual property directly.

3.8 United States law on embryonic stem cell research

A useful illustration of regulation by funding condition rather than by prohibition.

1. **1974**. Congress banned nearly all federally funded fetal tissue research until the National Commission devised guidelines.
2. **1975**. An Ethics Advisory Board was established to examine fetal and fetal tissue research originating from abortions.
3. **1980**. President Reagan disbanded the Board, producing a de facto moratorium on federal funding of human embryo research.

4. **1988.** A federal panel approved funding after lengthy debate.
5. **1990.** President George H. W. Bush vetoed the bill lifting the moratorium.
6. **1993 and 1994.** President Clinton lifted the moratorium by executive order, then reversed it after public pressure.
7. **1996.** The **Dickey Wicker Amendment**, named after Jay Dickey and Roger Wicker, was enacted as an appropriations rider prohibiting federal funding for the creation of human embryos for research and for research in which embryos are destroyed. *The lecture dates it to 1995; the correct year is 1996, and it has been re enacted annually since.*
8. **1998.** James Thomson isolated human embryonic stem cells through a process that destroyed embryos, which began the modern debate.
9. **2001.** President Bush prohibited federal funding, though the policy did not affect the private sector or State funded research.
10. **2005.** The National Academies released Guidelines for Human Embryonic Stem Cell Research.
11. **2007.** Shinya Yamanaka and James Thomson independently derived induced pluripotent stem cells.
12. **2009.** President Obama reversed the 2001 executive order.

Sherley v. Sebelius, United States Court of Appeals for the District of Columbia Circuit, 24 August 2012, reported at 689 F.3d 776

Facts. Two adult stem cell researchers challenged the National Institutes of Health guidelines implementing President Obama's executive order, arguing that federal funding of embryonic stem cell research violated the Dickey Wicker Amendment. An earlier round on a preliminary injunction is reported at 644 F.3d 388 (D.C. Cir. 2011).

Held. The court **upheld federal funding**. Applying the Chevron doctrine it held that the word **research** in the Amendment is ambiguous, and that the agency's interpretation permitting funding of work using cells derived from embryos destroyed in an earlier, separately funded step was **reasonable**. Summary judgment for the government was affirmed.

Principle. **Where a statute is ambiguous, an agency's reasonable interpretation governs, even on a question of deep moral contestation.** The leading illustration of bioethics being resolved through administrative law rather than constitutional law. *Verify the 689 F.3d citation before quoting it.*

3.9 Stem cell research and regulation in India

The threshold legal classification

1. **Stem cells and their derivatives fall within the definition of **drug** under the Drugs and Cosmetics Act, 1940**, and are categorised as an **Investigational New Drug** or **Investigational New Entity** when used for clinical application.
2. **The commercial use of stem cells as elements of therapy is prohibited.**
3. **No stem cell administration to humans is permissible outside the purview of clinical trials.**

This classification is what gives the regulator jurisdiction. Because a stem cell is a drug, the Central Drugs Standard Control Organisation can license it, inspect the facility and prosecute misuse, **so the unproven stem cell therapy clinic is legally an unlicensed drug manufacturer and not merely an unethical practitioner.**

The guidelines and the institutions

1. **Guidelines for Stem Cell Research and Therapy, 2007.**
2. **National Guidelines for Stem Cell Research, 2013.**
3. **National Guidelines for Stem Cell Research, 2017**, issued jointly by the Indian Council of Medical Research and the Department of Biotechnology on **11 October 2017**, whose guiding philosophy is to **prevent premature commercialisation of unproven stem cell therapies** while generating knowledge on a sound scientific rationale.

A **Cell Biology Based Therapeutic Drug Evaluation Committee** was constituted in 2010 by the drug regulator to review cell therapy trials, and the Apex Committee of the regulator later recommended that proposals evaluated by it be placed **directly before the Apex Committee** without passing through the Technical Committee.

Under the 2013 and 2017 guidelines every institution carrying out research on human stem cells had to constitute an **Institutional Committee for Stem Cell Research** and register it with the **National Apex Committee for Stem Cell Research and Therapy**. The institutional committee approved and monitored research at institution level; the national committee monitored nationally and laid down guidelines.

Important current correction. That two tier structure was dissolved in March 2024, citing difficulties in implementation. Review of stem cell research now sits with **Institutional Ethics Committees** operating within the framework of the National Ethical Guidelines of 2017. The 2017 stem cell guidelines otherwise remain the reference document, and commentators note that the redistribution of the dissolved committees' functions has not been fully clarified.

Scope of the 2017 guidelines

1. They **do not apply** to research using non human stem cells and their derivatives.
2. They **do not apply** to haematopoietic stem cells used for haematological, immunological and metabolic disorders, those being an established standard of medical care.
3. **Platelet rich plasma and autologous chondrocyte or osteocyte implantation** fall outside, being other cell based applications rather than stem cell transplantation.
4. Unique provisions are justified by **unlimited proliferation, differentiation into cells of the germ layers, oncogenic potential, unrecognised toxicities, and possible involvement in preimplantation stages of human development.**

Ethical and scientific requirements

On the **donor**: informed consent is mandatory before procurement and includes **video consent** under the audio visual recording guidelines of January 2014. Confidentiality is sacrosanct but provision must exist for **traceability in a contingency**. The donor must be told about screening for transmittable diseases and about procedural risks of collection under anaesthesia, and must be informed that cell lines or products may be generated, banked, shared, genetically manipulated and commercialised, and that **the intellectual property rights in the biological material will not vest with the donor**, though efforts should be made to pass on any financial benefit to the donor or community. The option of sharing intellectual property must be stated in the consent form.

On **manufacture and quality**: processing must occur in a facility **licensed by the drug regulator and compliant with good manufacturing practice**, and quality control must comply with **Schedule M** of the Drugs and Cosmetics Act. Pluripotent cells carry added risks, being able to **acquire mutations in prolonged culture, differentiate into inappropriate phenotypes, form benign teratoma or malignant outgrowths, and fail to mature**. Release criteria cover **cell viability, final cell population using cell surface markers, stability and release requirements**, with labelling, standard operating procedures, certification of the facility, and submission of chemistry, manufacturing and control documents.

On **evidence**: there is a lack of solid scientific evidence of clinical efficacy other than haematopoietic transplantation for approved indications. Protocols need defined primary and secondary endpoints and **follow up of at least two years, and a participant must not be charged for any procedure related to the trial, including hospital stay and laboratory investigations.**

Levels of manipulation and the permissions required

1. **Minimal manipulation, homologous use for unapproved indications.** Approvals of the institutional stem cell committee, the ethics committee and the drug regulator. Example: mononuclear cells from bone marrow, peripheral blood or umbilical cord blood given intravenously.

2. **Minimal manipulation, non homologous use.** Drug regulator approval is mandatory in addition. Example: the same cells by any route other than intravenous for neurological, musculoskeletal, liver or cardiovascular disorders.
3. **Same surgical procedure exception.** Cells removed and implanted into the same individual during one operation should not undergo processing beyond centrifugation, rinsing, cleaning or sizing.
4. **Substantial manipulation.** Drug regulator approval only after institutional stem cell and ethics clearances. Example: isolation of the stromal vascular fraction from adipose tissue by enzymatic digestion or mechanical disruption, with or without culture.
5. **Major manipulation.** Genetic and epigenetic modification, transient or permanent, altering not only numbers but biological characteristics and function, including transdifferentiation, transduction or transfection by retroviral or lentiviral vectors, and regulated lineage specific differentiation of embryonic and induced pluripotent cells.

Permissible, restrictive and prohibited research

Permissible, with prior approval of the institutional stem cell and ethics committees: in vitro studies using stem cells isolated from tissues; establishment of new embryonic lines from spare embryos or induced pluripotent lines from fetal or adult somatic cells, subject to consent, registration, deposit in an accredited cell bank and a unique identification number; in vitro studies using established lines; and in vivo studies in animals other than primates, with the animal ethics committee's approval, provided such animals are **not allowed to breed** if the cells are likely to be incorporated in the gonads.

Restrictive, requiring extra justification or higher approval:

1. **Creation of human preimplantation embryos** by in vitro fertilisation, intracytoplasmic sperm injection, somatic cell nuclear transfer or otherwise, specifically to derive cell lines. The investigator must show the research cannot be done with existing or spare embryo derived lines, define the minimum number of embryos, and have appropriate expertise.
2. **Trials sponsored by multinationals using products developed outside India**, needing clearance from the regulator of the country of origin and prior Indian approval after institutional clearances. International collaborations also need the **Health Ministry's Screening Committee**, and import needs a **licence from the drug regulator**.
3. **Introduction of human cells into animals, including primates**, at embryonic or fetal stages. If human cells could contribute in a major way to the **brain or gonads**, justification must be substantiated with data, the animals must not breed, and higher approval was required.
4. **Chimera studies** mixing stem cells from two or more species at any stage of early development.
5. **Genome modification, including gene editing by CRISPR Cas9, of stem cells, germ line cells, gametes or embryos**, which is **restricted to in vitro studies only** and requires review by the institutional stem cell, ethics and biosafety committees and finally by the **Review Committee on Genetic Manipulation**. Only spare embryos, germ line cells or gametes may be used, and **genome modified human embryos must not be cultured beyond fourteen days of fertilisation or the formation of the primitive streak, whichever is earlier**.

Prohibited research, a list of seven to be memorised as a list.

1. **Human germ line gene therapy and reproductive cloning.**
2. **In vitro culture of intact human embryos beyond fourteen days** of fertilisation or formation of the primitive streak, whichever is earlier.
3. **Clinical trials involving xenogeneic cells.**
4. **Any clinical research on xenogeneic and human hybrids.**
5. **Use of genome modified human embryos, germ line stem cells or gametes for developmental propagation.**
6. **Implantation of human embryos generated by any means, after in vitro manipulation, at any stage, into the uterus in humans or primates.**

7. **Breeding of animals in which human stem cells have been introduced at any stage and are likely to contribute to chimeric gonadal cells.**

The pattern across all three lists is the examinable point. **Indian stem cell regulation draws its lines at three places: the fourteen day limit, the germ line, and the species boundary.** Every prohibition applies one of those three.

3.10 Where bioethics is codified in Indian statute

Bioethics in India is not only guidance. Six statutes convert particular bioethical judgements into binding law, and naming them shows breadth in an answer. Confirm the amendment years before citing them precisely.

1. **Transplantation of Human Organs and Tissues Act, 1994**, amended in 2011, on organ donation, brain death and the prohibition of commercial dealing in organs.
2. **Medical Termination of Pregnancy Act, 1971**, amended in 2021, on lawful abortion.
3. **Pre Conception and Pre Natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994**, as amended in 2002, which is the statute that constrains preimplantation and prenatal diagnosis.
4. **Assisted Reproductive Technology (Regulation) Act, 2021.**
5. **Surrogacy (Regulation) Act, 2021.**
6. **Mental Healthcare Act, 2017**, on capacity, advance directives and consent to treatment.

To these add the **Drugs and Cosmetics Act, 1940 with the New Drugs and Clinical Trials Rules, 2019** for research on human participants, and the **Prevention of Cruelty to Animals Act, 1960** for animals.

3.11 Bioethics and patent law

Bioethics enters the Patents Act, 1970 through **section 3(b)**, which excludes inventions whose primary or intended use or commercial exploitation would be **contrary to public order or morality**, or would cause serious prejudice to human, animal or plant life or health or to the environment. This is the domestic counterpart of Article 27.2 of the TRIPS Agreement, and it is the provision under which a biological weapon, a human cloning process or a terminator seed would be refused on moral grounds. **Do not confuse section 3(b), the morality clause, with section 3(d), which concerns new forms of known substances.**

PART 4 REGULATORY FRAMEWORK FOR GM CROPS: INDIA, UNITED STATES, EUROPEAN UNION

4.1 Why plant biotechnology at all

The case begins with a demographic fact: **exponential growth in population raises food demand while farming land shrinks**. Plant biotechnology promises **high yield in low time** and **resistance to environmental hazards**. Transgenic plants are developed for **insect resistance, herbicide tolerance, disease resistance, high nutritional quality and yield potential, delayed fruit ripening, and enhanced ornamental value**, and plants are also used as **bioreactors** for biopolymers, nutraceuticals, antigens, vaccines and monoclonal antibodies.

The frequently cited meta analysis by Klümper and Qaim (2014) concluded that transgenic technology **reduced pesticide use by 37 per cent, increased crop yields by 22 per cent and increased farmer profits by 68 per cent**. Present these as the findings of that study rather than as settled fact, because the effect sizes are contested.

The commercial landscape

1. **Calgene's FLAVR SAVR tomato**, with delayed premature fruit softening, was approved for commercial release by the United States Food and Drug Administration in **1994**. It was the first.

2. The first transgenic plant accepted and approved for use in the **European Union in 1998 was Corn MON 810**, developed by Monsanto.
3. In **2014** genetically modified crops were grown by **18 million farmers in 28 countries** on **181.5 million hectares**, about **13 per cent of the world's arable land**.
4. **Four crops, corn, soy, cotton and canola, account for 99 per cent of global cultivation.**
5. The five fast developing producers are **China and India in Asia, Argentina and Brazil in Latin America, and South Africa**, together growing 71.4 million hectares, about 44 per cent of global acreage.

Traits by crop: abiotic stress tolerance in soybean, maize and sugarcane; altered growth and yield in soybean and maize; herbicide tolerance in corn, soybean, cotton, canola, sugar beet, rice and flax; virus resistance in papaya, squash and potato; altered oil composition in canola and soybean; delayed fruit ripening in tomato; male sterility and restorer systems in chicory and corn; and insect resistance in corn, cotton, potato and tomato.

4.2 Public fear, examined honestly

Human and animal health

1. A **1998 finding** that genetically modified potatoes expressing a gene for snowdrop sugar binding protein caused **stunted growth in rats and compromised their immune systems**. The study is contested but it is the origin of much public concern.
2. **Allergenicity**. A **1996 report of allergenic reaction to transgenic soybean**, where Brazil nut albumin genes led to high levels of methionine. Against that, transgenic rice carrying the entire coding sequence of the soybean ferritin gene was found to pose **no allergenicity risk**. **Allergenicity is therefore a real but product specific risk, which is precisely why regulation is case by case.**
3. **Antibiotic resistance**, through apprehended horizontal transfer of antibiotic resistance marker genes to human gut flora.
4. **Conversion of normal food into metabolically dangerous food**, and concern about the presence of **foreign DNA** and of the **vector and vector elements** in food.

Environmental effect

1. **Danger to non target insects**. Pollen from Bt corn was found toxic to the **monarch butterfly** (Losey and others, 1999), which aggravated public concern more than any other single finding.
2. Insect resistant crops may harm **beneficial insect predators** such as ladybird beetles (Birch and others, 1997) and lacewings (Hilbeck and others, 1998).
3. **Development of super plants and super weeds** through gene flow.

Socioeconomic and other concerns

Patenting and monopolisation, and **high seed price**, then **biosafety issues** and **moral perplexity**, of which the example given is the **terminator gene**. The last is where agricultural biotechnology becomes a question of ethics rather than of risk assessment.

On terminator technology, Indian law is precise and it is better to cite the provisions than to assert a ban. Under **section 18(1)(c)** of the Protection of Plant Varieties and Farmers' Rights Act, 2001 every application for registration must carry an **affidavit that the variety contains no gene or gene sequence involving terminator technology**, and under **section 29(3)** no variety involving technology injurious to the life or health of humans, animals or plants may be registered, the Explanation stating that **any technology includes genetic use restriction technology and terminator technology**. **Note the limit: the Act denies registration; it is not itself a general prohibition on developing or importing such a variety.**

4.3 The three regulatory philosophies

1. **United States: product based and market facilitating.** Regulation focuses on the **characteristics of the end product**. Existing statutes are deemed sufficient. **The burden of proof lies with the regulator to show harm.**
2. **European Union: process based and precaution oriented.** **The method of creation triggers regulatory review.** Safety must be demonstrated before approval, so the burden lies on the applicant. Consumer trust is prioritised.
3. **India: hybrid, evolving and reactive.** A mixed jurisdiction system **anchored in environmental law** but increasingly pressured by food safety mandates. Multi agency, politically sensitive, and struggling to move from reactive crisis management to proactive policy.

4.4 United States

The Coordinated Framework, 1986

Established in the same year as India's Environment (Protection) Act but on the opposite philosophy. It decreed that **existing statutory authority was sufficient**, and that regulation should focus on the characteristics of the end product, not the process. Three agencies divide the work.

1. **USDA APHIS** evaluates potential **plant pest risk** and environmental safety before field trials and commercialisation.
2. **EPA** regulates crops with **pesticidal properties**, such as Bt crops, including tolerance levels and resistance management.
3. **FDA** oversees **food and feed safety** of derived products under the Federal Food, Drug and Cosmetic Act.

The FDA position and the GRAS standard

The landmark 1992 Policy Statement concluded that genetically modified foods are not inherently different from their conventional counterparts, and therefore do not require special premarket approval, only voluntary consultation. The operative standard is **generally recognised as safe**: if composition does not materially differ from the conventional counterpart in ways that raise safety concerns, the product is presumed safe. **This shifts the burden, so the product must be proven unsafe to be blocked, not proven safe to be approved.** The regulator examines **nutritional profile, potential allergenicity and compositional equivalence**, and the method of modification is legally irrelevant.

The SECURE Rule, 2020

The 1987 APHIS regulations were built around a **plant pest risk** trigger that many modern innovations did not fit. The Sustainable, Ecological, Consistent, Uniform, Responsible, Efficient Rule modernised oversight by creating **categorical exemptions for crops that could have been developed through conventional breeding**, streamlining the pathway for genome edited crops. The legal insight worth quoting: **risk proportionate regulation is not the same as deregulation; applying less scrutiny to low risk innovations frees regulatory resources for genuine novel risks.**

The anti browning mushroom, USDA APHIS decision, letter dated 13 April 2016

Facts. A researcher at Pennsylvania State University developed a white button mushroom, *Agaricus bisporus*, edited with CRISPR Cas9 to suppress browning by small deletions in a polyphenol oxidase gene. **No foreign DNA was inserted.** The request for confirmation was made on 30 October 2015.

Held. APHIS determined it **lacked regulatory authority** because the mushroom contained no plant pest components, the traditional trigger for oversight, and declared it to have **non regulated status**. It was the first CRISPR Cas9 edited crop deemed to need no United States review, and within a week the agency reached the same conclusion for an edited corn variety.

Principle. **Precise gene editing can lawfully bypass a trigger written for transgenic technology.** The case is cited worldwide, including in India's deliberations, as evidence that a lighter touch approach is legally defensible. *Correction: the developer was Yinong Yang, not the name given in the course slides.*

4.5 European Union

The European Union is the most cautious major jurisdiction. Its approach is rooted in the **precautionary principle, codified in Article 191 of the Treaty on the Functioning of the European Union**, and demands rigorous assessment before authorisation, **placing the burden of proof on the applicant to demonstrate safety rather than on the regulator to demonstrate harm.**

The four instruments

1. **Directive 2001/18/EC** on deliberate release: authorisation, environmental risk assessment, case by case and step by step evaluation, monitoring and time limited consents.
2. **Regulation (EC) 1829/2003** on genetically modified food and feed, the centralised framework. **Unlike the United States, the European Union created dedicated GMO specific legislation, and the method of creation is itself the trigger regardless of whether the final product differs from conventional varieties.**
3. **Regulation (EC) 1831/2003** on labelling and traceability.
4. **Directive (EU) 2015/412**, permitting a Member State to **opt out of cultivation** of a Union authorised product on grounds other than risk.

EFSA and the single door procedure

The **European Food Safety Authority** conducts independent risk assessment through its Panel on Genetically Modified Organisms, evaluating **molecular characterisation, comparative safety, toxicology, allergenicity and environmental impact. The Authority advises; the final authorisation decision rests with the European Commission and the Member States politically, so a positive scientific opinion does not guarantee authorisation.** The single door runs in four steps.

1. The applicant submits to the **national authority of one Member State**, which forwards it to the food safety authority.
2. The authority conducts an **independent scientific risk assessment** and issues an opinion.
3. The **Commission prepares a draft decision** based on that opinion.
4. The **Member States vote**, giving political approval or rejection.

Transparency and traceability are legally mandated at every step, which gives the system public credibility at the cost of speed and flexibility.

Labelling, the 0.9 per cent threshold and zero tolerance

All food and feed containing **more than 0.9 per cent of authorised genetically modified content** must be labelled; below that, accidental contamination is presumed and labelling is not required. Every product carries a **unique identifier throughout the supply chain**, enabling traceability from farm to fork, which serves both **product recall in a safety emergency** and **informed consumer choice**. For a lawyer, traceability documentation is both a compliance obligation and an evidentiary tool in product liability disputes.

Modern techniques detect traces far below any level posing risk, which creates a paradox: **a theoretically achievable zero tolerance standard is unworkable in a global supply chain, since a single dust particle from an approved crop could render a shipment non compliant.** The 0.9 per cent threshold answers this for **authorised** events. For **unauthorised** events the European Union maintains a de facto **zero tolerance**, creating enormous exposure for bulk commodity traders. India has not established equivalent clarity in its import tolerance policy.

The 2018 ruling and the reform that followed

Confédération paysanne v. Premier ministre, Court of Justice of the European Union, Grand Chamber, 25 July 2018

Facts. The French Conseil d'Etat referred the question whether organisms produced by new mutagenesis techniques, including CRISPR gene editing, fall within the definition of a genetically modified organism under Directive 2001/18/EC.

Held. They do. The risks associated with the new techniques **were potentially comparable to those of traditional transgenics** and therefore warranted equal scrutiny. The exemption in Annex I B covers only mutagenesis techniques **conventionally used with a long safety record**. Advocate General Bobek had advised the opposite.

Principle. All CRISPR edited plants became trapped in 2001 era legislation requiring full authorisation lasting years. The scientific community criticised the ruling for drawing **no distinction between random chemical mutagenesis, which is exempt, and precise gene editing, which is regulated**. **The innovation impact was immediate: research programmes relocated outside the European Union and European scientists increasingly conduct field work in the United States, Australia and Japan.** Applied by the Conseil d'Etat in February 2020.

In a reversal, the European Union has been advancing a **New Genomic Techniques Regulation** expected in the 2025 and 2026 period, creating two categories. **NGT 1 plants**, being those that could have occurred naturally or through conventional breeding, are treated as conventional and exempted from Directive 2001/18. **NGT 2 plants**, with more complex modifications, remain under full authorisation. **This is the European Union's clearest acknowledgment that its 2001 framework was not designed for modern precision biotechnology.** Confirm final adoption before describing it as law.

4.6 The trade dimension: the World Trade Organization dispute

European Communities, Measures Affecting the Approval and Marketing of Biotech Products, WT/DS291/R and connected reports, adopted 21 November 2006

Facts. The United States, Canada and Argentina complained that the European Communities had stopped approving biotech products, delayed individual applications, and permitted national bans on already approved products. The panel was established on 29 August 2003 and the reports circulated on 29 September 2006. There was no appeal.

Held. Three findings. A **general de facto moratorium** existed from **June 1999 to August 2003**, which the European Communities had denied; it was not itself a sanitary measure but caused failure to complete approval procedures **without undue delay**, contrary to **Annex C(1)(a) and Article 8 of the SPS Agreement**. There was undue delay in **twenty four of twenty seven** products. Nine **national safeguard measures** breached **Articles 5.1 and 2.2** because they rested on no risk assessment and could not be saved as provisional measures under Article 5.7.

Principle. **A Member may regulate genetically modified products strictly, but it may not simply refuse to decide.** The panel did not rule that such products are safe, and did not rule on the European approval legislation itself, which was never challenged; it ruled on **procedural delay** and on the **evidentiary basis of national bans**. India is exposed in the same way: an indefinite moratorium with no formal, reasoned, science based refusal is legally weaker than an express refusal.

4.7 The comparison, in four dimensions

1. **Legal basis.** United States: the Coordinated Framework of 1986 using existing statutes. European Union: dedicated legislation, Regulation 1829/2003. India: the Environment (Protection) Act, 1986, the Rules, 1989 and the Food Safety and Standards Act, 2006.
2. **Authorisation type.** United States: product based, case by case, with categorical exemptions. European Union: centralised, process triggered, politically finalised. India: multi agency, evolving, **with no single window clearance.**
3. **Key scientific body.** United States: FDA, USDA APHIS and EPA, distributed. European Union: the food safety authority, centralised and independent. India: GEAC, RCGM and FSSAI, overlapping.
4. **Approval speed.** United States: relatively fast, typically twelve to twenty four months. European Union: slow, often four to seven years or more including the political stage. India: **highly variable and indefinitely stalled for food crops.**

4.8 Stacked events

A **stacked event** is a crop combining multiple approved traits, for example herbicide tolerance and insect resistance, through conventional cross breeding of two separately approved parents. Treatment differs sharply.

1. **European Union.** Each new combination is a **new organism requiring full independent authorisation**, so the same traits are assessed repeatedly.
2. **United States.** Stacked products from previously deregulated parents are **generally not subject to new review** if no novel risks arise, a position the SECURE Rule clarified further.
3. **India.** A **full assessment is required for each new genetic combination**, even where the component traits were individually approved.

The Indian consequence is concrete. **A farmer with access to a herbicide tolerant crop and to an insect resistant crop cannot easily obtain a variety combining both without triggering an entirely new review cycle**, which adds years and cost and insulates the domestic market from the multi trait varieties standard elsewhere.

4.9 Labelling, stigma and the Indian gap

1. **The right to know as an autonomy right.** The European regime rests on the principle that consumers may know what is in their food **regardless of whether it poses a safety risk**, which makes labelling function like allergen labelling: disclosure irrespective of harm.
2. **The stigma effect.** Mandatory labelling has empowered the non genetically modified marketing industry, and products labelled GM free command premium prices, so **the existence of the label implies risk even where regulators have determined that none exists.**
3. **The Indian gap.** India lacks a comprehensive enforced labelling framework. The food authority has proposed regulations but implementation has been repeatedly delayed, creating uncertainty for businesses and **arguably depriving consumers of information they may have a right to access under Article 21 read with the right to health and the right to information.** That is a live constitutional argument, not a settled one.

4.10 Harmonisation and trade risk

1. **Codex Alimentarius.** A joint FAO and WHO body with internationally recognised standards for GM food safety assessment. Though **not legally binding, Codex standards are referenced in World Trade Organization dispute settlement** as benchmarks for science based regulation, and **India's alignment is only partial, which creates vulnerability in trade disputes.**
2. **OECD BioTrack.** A database of national regulatory decisions; greater use of it and mutual recognition agreements could reduce duplication.
3. **Asynchronous approvals.** Where a trait is approved in one jurisdiction and not another, **a single unapproved event detected in a bulk shipment can trigger rejection of entire cargoes worth millions**, so regulatory fragmentation is a major source of international trade risk.

Part 4A Applicable Regulations on GM Crops in India: EPA 1986, Rules 1989 and GEAC

4A.1 The Environment (Protection) Act, 1986 as the legislative bedrock

The Environment (Protection) Act, 1986 is the constitutional spine of India's GM regulatory system. It was enacted in the wake of the **Bhopal disaster** and gave the Central Government sweeping powers to take measures protecting the environment, a mandate broad enough to encompass biotechnology risks decades before they became a policy concern.

Under **section 6** the Government made the **Rules for the Manufacture, Use, Import, Export and Storage of Hazardous Micro Organisms, Genetically Engineered Organisms or Cells, 1989**, the primary operative mechanism for GM governance in India, with **sections 8 and 25** supplying supporting powers.

Two structural observations for any answer. **India regulates agriculture through an environmental statute, which is why the environment ministry and not the agriculture ministry holds the approval power.** And the Act predates the technology it now governs, so its application to modern techniques depends on interpretation rather than on express words.

4A.2 Scope of the Rules, 1989

The Rules cover five categories of activity.

1. **Research and development in contained environments.**
2. **Large scale use** of genetically engineered organisms.
3. **Import and export** of such organisms and products.
4. **Environmental release and commercial production.**
5. **Transportation** of hazardous biological material.

They apply not only to the organisms, micro organisms and cells but to **any substances, products and foodstuffs of which such cells, organisms or tissues form part, so the Rules follow the organism into the finished product.**

A body of guidance sits under the Rules and is worth naming: the **Recombinant DNA Safety Guidelines, 1990**; guidelines of 1999 for generating preclinical and clinical data for recombinant vaccines, diagnostics and other biologicals; the **Guidelines and Handbook for Institutional Biosafety Committees, 2011**; and guidance on confined field trials, food safety assessment and environmental risk assessment.

4A.3 The six authority web

India distributes responsibility across six bodies with specific but often overlapping mandates. **Many notes wrongly record four; the correct number is six.** Learn the function label attached to each.

1. **RDAC, the Recombinant DNA Advisory Committee.** In the Department of Biotechnology. Purely **advisory**; the scientific advisory backbone providing policy guidance and risk frameworks, reviewing developments in India and abroad and recommending appropriate safety regulations.
2. **IBSC, the Institutional Biosafety Committee.** The **first point of oversight**, embedded in each research institution and laboratory.
3. **RCGM, the Review Committee on Genetic Manipulation.** In the Department of Biotechnology. **Monitors contained research**, clears small scale and **confined field trials**, and issues manuals of guidelines and standard operating procedures.
4. **GEAC, the Genetic Engineering Appraisal Committee.** **The apex nodal body under the environment ministry for final environmental approval**, covering large scale use and **release of organisms and products into the environment**, including experimental field trials, with power to act immediately in case of hazard.
5. **SBCC, the State Biotechnology Coordination Committee.** **Monitoring** at State level, with power to inspect, investigate and take punitive action through the State Pollution Control Board.
6. **DLC, the District Level Committee.** **Monitoring** at district and installation level, investigating compliance and reporting upward.

Memory device: advisory, institutional, research monitoring, approval, and two monitoring tiers.

4A.4 The role of the GEAC in detail

1. **Status.** Apex nodal body under the Ministry of Environment, Forest and Climate Change, constituted under **Rule 4(4)** of the Rules, 1989.

2. **Core function.** Approval of activities involving **large scale use** of hazardous micro organisms and recombinants in research and industrial production from the environmental angle, and approval of the **release of genetically engineered organisms and products into the environment**, including experimental field trials.
3. **Prohibitory effect.** **Nothing containing a genetically engineered organism may be produced, sold, imported or used without its permission.**
4. **Emergency power.** It may take **immediate action in the case of a hazard.**
5. **The change of name.** It was originally the Genetic Engineering **Approval** Committee and was renamed the Genetic Engineering **Appraisal** Committee in 2010, following the Bt brinjal moratorium. **The renaming was a deliberate signal that the technical body appraises and the political executive decides.**
6. **The limit of its power.** A clearance from the committee is a permission, not an entitlement. The environment minister may override it, and State governments may block implementation.

4A.5 The Institutional Biosafety Committee in depth

This body deserves separate treatment because it is where regulation actually bites or fails. **Every institution conducting contained GM research must have a functioning committee,** composed of **scientists, biosafety experts, a medical officer and nominees of the Department of Biotechnology.** It reviews and approves experimental protocols, classifies risk and containment, maintains records, prepares the **on site emergency plan**, and reports to the research committee.

The practical criticism is examinable. **The committees are the critical link between high level policy and day to day compliance, yet their effectiveness varies enormously: well resourced institutes maintain robust functions while smaller institutions often treat the requirement as a paperwork exercise.** For a lawyer this implementation gap is both a compliance risk for the client and a systemic failure that undermines the credibility of the whole architecture.

4A.6 The approval pathway in sequence

1. **Institutional Biosafety Committee approval** for contained laboratory research, fixing the risk group and containment level. A soil bacterium can be handled in an ordinary laboratory; a tuberculosis organism or a virus of the COVID class needs high containment.
2. **RCGM clearance** for confined and then multi location field trials, which generate the biosafety dossier.
3. **GEAC approval** for environmental release, on appraisal of the complete dossier.
4. **FSSAI food safety clearance** for commercial sale of genetically modified food.
5. **Monitoring** thereafter by the State and district committees, with State governments deciding operational aspects of release in their territory.

Each stage has its own timelines, documentation and political sensitivities, and failure at any stage can require restarting from an earlier approval. Regulatory mapping, meaning knowing which agency must act before another can, is itself a core legal service.

4A.7 The GEAC and FSSAI fault line

The **Food Safety and Standards Act, 2006** created the Food Safety and Standards Authority as an independent statutory regulator for food, including food derived from genetically modified organisms. **This created a structural ambiguity: the appraisal committee governs environmental release under the 1986 Act, while the food authority claims jurisdiction over food safety for human consumption under section 22 of the 2006 Act.**

Correction to the course slides. They describe a **Supreme Court mandate of 2017** directing that genetically modified food be governed under the food safety framework. What the record shows is an **affidavit filed by the food authority in the Supreme Court in May 2017**, stating that the appraisal committee is responsible for risk assessment and approval of release into the environment, and that once any genetically modified food

is approved by that committee, the food authority is responsible under section 22 for regulating its manufacture, distribution, sale and import. **Cite it as the food authority's stated position in an affidavit, not as a judicial direction.**

The consequence is a genuine administrative law problem. Neither regulator has clear supremacy. An applicant who clears the environmental process enters a **parallel, poorly coordinated review with no clear timelines, no single window and no legal obligation on either agency to defer to the other. This structural stalemate has effectively created an informal moratorium on non cotton GM crops without any formal legal basis.** That is the strongest single criticism available in this topic.

Two related current facts. The food authority required a **non genetically modified and GM free certificate** for a list of imported food crops from **1 January 2021**, later clarified as not extending to processed foods generally. Its **draft Genetically Modified or Engineered Foods Regulations** went through consultation in 2021 and 2022 and have not been finally notified, which is why India still lacks an operative labelling regime.

4A.8 Genome editing: the 2022 recalibration

In 2022 the environment ministry issued an **Office Memorandum** providing a framework for genome edited plants, described in the course material as **arguably the most important development in Indian GM regulation since the approval of Bt cotton in 2002.**

1. **SDN 1**, introducing only small insertions or deletions with **no template DNA**, and **SDN 2**, using a short DNA template for precise changes **within the organism's own genome**, are **exempted** from the strict pathway under the Rules, 1989.
2. **SDN 3**, which introduces **exogenous DNA**, remains fully regulated.
3. The institutional biosafety committee must certify the **absence of foreign DNA**.

For the first time India's framework distinguishes between the process of modification and the outcome, bringing it into alignment with the United States approach. This creates a potential fast track for climate resilient and nutritionally improved crops, and two genome edited rice lines were announced in May 2025. Two criticisms follow: India now applies a **product based test to editing and a process based test to transgenics**, which is doctrinally inconsistent; and it did so by **executive memorandum rather than by amending the Rules**.

4A.9 Five interpretive questions for a lawyer

The 1986 Act and the 1989 Rules were drafted before CRISPR, site directed nuclease techniques or synthetic biology existed as concepts. Courts and regulators therefore have significant room to interpret, and five questions are live.

1. Does **SDN 1 gene editing produce a genetically engineered organism** within the definition in the Rules?
2. Is the **2022 Office Memorandum legally binding or merely advisory?**
3. **Can a State government override a GEAC environmental clearance?**
4. **What standard of judicial review applies to GEAC decisions?**
5. Does the **Food Safety and Standards Act, 2006** impliedly repeal parts of the Rules, 1989?

These five make an excellent critique, because each identifies a defect without requiring you to assert a conclusion the law has not yet reached.

4A.10 Federalism, the current standoff and the 2024 directive

India's federal structure adds a second layer. **Even where the appraisal committee grants clearance, individual State governments, acting on their own political calculations, have blocked field trials and commercial cultivation.** The constitutional division of agricultural powers creates enforcement gaps that no central refinement can resolve without political consensus. The tension mirrors the European Union's relationship with its Member States, and resolving it requires either **constitutional clarification** or a **cooperative framework agreement**.

Three features define the present position.

1. **Research without reward.** India has a robust pipeline, including drought tolerant maize, fortified rice and pest resistant mustard, but commercial cultivation remains largely blocked. **The gap between research readiness and commercial approval is a regulatory failure with real economic and food security costs.**
2. **Political and civil society pressure.** Despite scientific clearances, food crops remain stalled, **revealing how political will often overrides scientific recommendation in India's federal structure.**
3. **The July 2024 Supreme Court directive**, requiring the Central Government to formulate a comprehensive national policy on genetically modified crops within a four month consultation window. It is both a judicial acknowledgment that the framework has failed and a political opportunity for reform.

The characterisation to carry into an essay: **India's GM governance has been reactive crisis management, with moratoriums issued in response to public pressure, court orders filling policy vacuums, and ministerial decisions substituting for statutory frameworks.**

Part 4B Bt Cotton and Bt Brinjal: Legal Controversies, State Bans and Liability

4B.1 Bt cotton: the numbers and why it worked

Bt cotton carries genes from *Bacillus thuringiensis* so that the plant makes a Cry protein toxic to bollworm larvae. It was approved for commercial cultivation in 2002 and **remains India's only commercially approved genetically modified crop.**

1. By 2014 it occupied approximately **95 per cent of India's cotton cultivation area**, one of the fastest and most complete agricultural technology adoptions in history.
2. **Pesticide use for cotton fell by an estimated 50 per cent** in major cotton States.
3. **Yields increased by 24 per cent on average** in the first decade.
4. India became the **world's largest cotton producer.**

The explanation for its success is not legal at all. **Bt cotton succeeded partly because cotton is a fibre crop, not a food; no one eats cotton directly, which reduced the emotional resistance that has blocked Bt brinjal and GM mustard.** This fibre against food distinction is **not written into India's law**, yet it has shaped regulatory and political decision making profoundly, and it illustrates how extra legal factors decide outcomes even where the science is settled.

The story also shows the limits of regulatory confidence. The approval was hard fought and contested by civil society, and was followed by continuing disputes over **royalty structures and trait efficacy**. It remains the template against which later applications are measured, and the reason food crops face a higher political and evidentiary bar.

4B.2 The legal controversies around Bt cotton

Price and trait value control

The **Cotton Seed Price (Control) Order, 2015**, issued under the Essential Commodities Act, 1955, fixes the **maximum sale price of Bt cotton seed** and the **trait value or licence fee** a technology provider may charge. **It converted a contractual relationship into a regulated one and is the direct origin of the litigation below.**

The patent and licensing dispute

Monsanto Technology LLC v. Nuziveedu Seeds Ltd., (2019) 3 SCC 381 (Supreme Court, 8 January 2019, Nariman and Navin Sinha JJ)

Facts. Monsanto held Indian **patent No. 214436**, claims 1 to 24 being process claims and claims 25 to 27 product claims to a **nucleic acid sequence**. A sublicense of 21 February 2004 allowed Indian seed

companies to develop hybrid Bt cotton seed against a trait value. After price control the licensees paid the statutory rate rather than the contractual rate, and Monsanto **terminated on 14 November 2015** and sued for patent and trade mark infringement. The defendants counterclaimed for revocation under **section 3(j)**, arguing that once inserted the sequence becomes an inseparable and inheritable part of the plant. Monsanto argued it was a man made laboratory chemical that could be removed again.

Held. The Supreme Court **set aside the Division Bench, restored the Single Judge's order of 28 March 2017 and remanded the suit**. Patentability and the section 3(j) exclusion are **mixed questions of law and fact requiring expert evidence and a trial**, and summary adjudication at the injunction stage was impermissible. All questions of fact and law were **expressly left open**, including whether the inserted sequence can be removed from the variety and whether the patented sequence is a plant or a part of a plant.

Principle. **India still has no authoritative ruling on whether a gene construct inserted into a plant is patentable.** For this Part the further point is that Monsanto's exposure arose not from any defect in its patent but from a **price control order** that made its contractual trait value unenforceable, which shows price regulation, contract, patent law and plant variety law interacting in a single dispute.

Illegal cultivation

Unapproved herbicide tolerant Bt cotton has been reported in cultivation in several States, and illegal Bt brinjal has been reported in Haryana and elsewhere. **Illegal cultivation is the clearest proof that the monitoring tier, the State and district committees, does not function**, and it creates a problem no approval framework can solve: the technology diffuses whether or not it is approved, and once it is in the field there is no practical way to recall it.

4B.3 Bt brinjal: the moratorium

The single most examinable narrative in this course.

1. **14 October 2009.** The Genetic Engineering Appraisal Committee recommended approval for environmental release of Bt brinjal, developed by Mahyco with Monsanto and public sector collaborators.
2. **January and February 2010.** The then Minister of Environment and Forests, Jairam Ramesh, held **public consultations in seven cities**, attended by several thousand people, and commissioned a scientific review.
3. **9 February 2010.** The Minister announced an **indefinite moratorium** on commercial release, expressly on a **precautionary principle based approach**, to last until independent scientific studies establish safety and long term impact on human health and the environment **to the satisfaction of the public and the scientific community**.
4. **17 February 2010.** He clarified that this was a **moratorium and not a permanent ban**, to remain until a political, scientific and societal consensus is reached.
5. **Thereafter.** The moratorium has never been lifted. The drug regulator and the Indian Council of Medical Research had asked for more exhaustive food safety testing, and major brinjal growing States had asked for deferment.

An institutional consequence often missed: it was in this period that the committee was renamed from **Approval to Appraisal**.

The two legal problems the moratorium exposes

1. **Scientific approval does not equal commercial release.** India's system gives significant discretionary power to the environment ministry, whose political calculus can override agency level scientific determination. **This political veto power has no clear legal standard and no time limit, which creates permanent regulatory uncertainty.** That is a classic administrative law objection: a discretion without a standard.
2. **The public consultation problem.** Who should have a voice: scientists, farmers, consumers or civil society? India's consultations have been criticised as poorly designed, **allowing organised opposition to dominate while the beneficiary farming communities remain underrepresented.** Process design is itself a legal and democratic question.

4B.4 The litigation: Aruna Rodrigues, the Technical Expert Committee and Gene Campaign

A long running public interest petition filed by **Aruna Rodrigues** challenged India's biotechnology regulatory system and sought a moratorium on release of genetically modified organisms pending a transparent and rigorous biosafety protocol conducted by independent expert bodies with published results. In **2012** the Supreme Court constituted a **Technical Expert Committee** with wide terms of reference. Its **2013 final report** was critical of the regulatory system, highlighted inadequacies and **inherent conflicts of interest**, and recommended a **ten year moratorium on commercial release of genetically modified food crops**. The regulatory regime was described in the later judgment as being in complete disarray.

The dispute came to a head over **genetically modified mustard, the hybrid DMH 11**, developed at Delhi University using the **barnase and barstar** system together with a herbicide tolerance gene. The appraisal committee recommended conditional approval for environmental release on **18 October 2022** and the ministry conveyed the decision on **25 October 2022**, which would have made it India's first transgenic food crop.

Gene Campaign and Another v. Union of India, 2024 INSC 545 (Supreme Court, 23 July 2024, B. V. Nagarathna and Sanjay Karol JJ)

Facts. Public interest petitions, heard with the Aruna Rodrigues petition and connected matters, challenged the approval for environmental release of DMH 11 and, more broadly, the adequacy of India's biosafety regulation.

Held. A **split verdict**. Justice Nagarathna **quashed** the approval: it violated statutory requirements, and where a statute prescribes a procedure it cannot be replaced by an alternative the statute does not contemplate; there was an **absence of an ICMR representative** at a crucial meeting, so health impacts went unconsidered; and the **Technical Expert Committee's recommendations were sidelined**. Justice Karol **upheld** it, holding the appraisal committee to be an expert body whose conditional approval showed no non application of mind, with field trials to continue under strict monitoring and the decision reviewable if circumstances changed.

Principle. Both judges agreed on five directions, which form the operative part of the case. **Because of the split the matter went to the Chief Justice for a larger Bench, so the approval stands neither quashed nor implemented and cultivation remains on hold.**

1. **Judicial review** of decisions on genetically modified organisms **is permissible**, so the regulator's scientific judgement is not immune.
2. The Union **must frame a national policy** on genetically modified crops covering research, cultivation, trade and commerce.
3. That policy must follow a **national consultation within four months** with all stakeholders including States and farmers.
4. Experts' credentials must be **verified scrupulously** and **conflicts of interest mitigated**, with rules framed for the purpose.
5. The **Food Safety and Standards Act** must be enforced on **imports of genetically modified food**.

Do not write that the Court banned genetically modified mustard, and do not write that it approved it. It split and referred the matter on.

4B.5 State bans and the federal question

Agriculture is a State subject, while environment and biotechnology are handled centrally and biosafety rests on a central environmental statute. Several States have refused no objection certificates for field trials or blocked cultivation after central clearance. Four legal points.

1. **Source of State power.** The State acts through its control over land, agriculture and public order, and through the monitoring role given to the **State Biotechnology Coordination Committee** under the Rules, 1989, which includes power to act through the State Pollution Control Board.
2. **The unresolved question**, already listed above: **can a State override a central environmental clearance?** There is no authoritative Indian ruling.
3. **The practical effect** is that a central approval is a permission, not an entitlement, and the applicant must secure political acquiescence in every State where the crop is to be grown.
4. **The comparative answer.** Directive (EU) 2015/412 expressly allows a Member State to opt out of cultivation of a Union authorised product on grounds other than risk. **The European Union solved by legislation the problem India has left to politics.** That is the sharpest reform proposal available in this topic.

4B.6 Liability: what the law provides and what it does not

This is the weakest part of the Indian framework and the place to distinguish yourself, provided you are honest about the gaps. Set it out in seven layers.

1. **Penal liability under the parent statute.** Contravention of the Rules, 1989 is a contravention of the Environment (Protection) Act, 1986, attracting the penalties in **section 15**, with **section 16** for offences by companies and **section 17** for offences by government departments. This is punitive, not compensatory.
2. **Regulatory liability.** The appraisal committee may take **immediate action in case of hazard**; the monitoring committees may inspect, investigate and act through the pollution control machinery; and approvals may be withdrawn.
3. **Civil liability in tort.** In the absence of a statutory scheme an injured farmer or neighbour must rely on negligence, nuisance, or the rule of strict and absolute liability developed in Indian environmental jurisprudence. **There is no reported Indian judgment awarding damages for harm caused by a genetically modified organism, so this remains theoretical.**
4. **Product liability.** A defective seed may attract remedies under consumer protection law and the Seeds Act framework, and for a registered variety **section 39(2) of the Protection of Plant Varieties and Farmers' Rights Act, 2001** imposes a duty on the breeder to disclose expected performance and to compensate the farmer if the propagating material fails to perform. Note that this is a **performance guarantee, not a biosafety liability**.
5. **Environmental compensation.** The National Green Tribunal Act, 2010 permits claims for compensation for damage arising from a breach of the Environment (Protection) Act, which is the most plausible statutory route for a biosafety claim.
6. **International liability.** The **Nagoya Kuala Lumpur Supplementary Protocol on Liability and Redress**, supplementing the Cartagena Protocol, adopted in 2010 and in force from 2018, requires response measures where damage to biological diversity is caused by a living modified organism, operating administratively: the competent authority requires the operator to act, or acts and recovers the cost. **India has no domestic legislation specifically implementing this liability regime.**
7. **Contractual liability.** In practice risk is allocated by contract: the technology licence, the sublicense, the material transfer agreement and the seed sale terms. That is why the Monsanto dispute was fought as a contract and patent case rather than as a biosafety case.

The conclusion for an answer: **India has a permissioning regime, not a liability regime. The 1989 Rules tell you who may do what; they say almost nothing about who pays when something goes wrong.** Both the Technical Expert Committee and the 2024 judgment point at this gap, and a national GM policy that does not create a liability and redress mechanism will have solved only half the problem.

4B.7 How to answer a question on this sub topic

Structure: the approval story for each crop; the legal controversies, being price control and patents for cotton and the moratorium and litigation for brinjal; the federalism problem and State bans; then liability in the seven layers; then a reform proposal. Take a position. The defensible one is that **a moratorium without a time limit**

and without a liability regime is the worst of both worlds: it neither permits the technology nor protects anyone from it.

PART 5 THE CARTAGENA PROTOCOL ON BIOSAFETY

5.1 Origin and status

1. **Parent instrument.** The Convention on Biological Diversity, 1992, a product of the Rio Earth Summit.
2. **Legal genesis.** Under **Article 19(3)** of the Convention, the Conference of the Parties established an Open ended Ad Hoc Working Group on Biosafety to develop a draft protocol focusing on the transboundary movement of any living modified organism resulting from modern biotechnology that may have an adverse effect on the conservation and sustainable use of biological diversity.
3. **Adopted 29 January 2000. Entered into force 11 September 2003.** India has been a party since 2003.
4. **Objective.** The safe transfer, handling and use of living modified organisms.

A definitional point that carries marks: a **living modified organism** is narrower than a genetically modified organism, because it must be **alive and capable of transferring or replicating genetic material**.

Processed food made from such an organism is not itself a living modified organism.

5.2 Scope, the three categories and the two exemptions

The Protocol applies to the **transboundary movement, transit, handling and use** of all living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity, **taking also into account risks to human health**.

1. **Living modified organisms for direct use as food or feed, or for processing.** Example: importing genetically modified soya to crush for oil.
2. **Living modified organisms destined for contained use.** Example: a bacterial or viral strain imported for laboratory research.
3. **Living modified organisms for intentional introduction into the environment.** Example: seed for cultivation.
4. **Exempt: pharmaceuticals for humans** addressed by other relevant international agreements or organisations.
5. **Exempt: products derived from living modified organisms** which no longer contain a living modified organism.

The illustration of why contained use matters is worth reproducing. A researcher who wants a sample of a dangerous pathogen from a foreign laboratory cannot simply carry it in a test tube. **The Protocol supplies notification, documentation, containment and liability, so that if the consignment breaks in transit there is a legal answer to the question of who is responsible.**

5.3 Advance Informed Agreement

The spine of the Protocol. Articles 8 to 10 and 12, subject to Articles 5 and 6, apply **prior to the first intentional transboundary movement of a living modified organism for intentional introduction into the environment** of the Party of import. Four components.

1. **Notification by the exporter**, being a detailed written description of the organism, given well in advance of the first shipment.
2. **Acknowledgement of receipt of the notification by the importer.**
3. **Decision procedure:** approve, prohibit, impose conditions, or request further information.
4. **Review of decisions**, including at the request of the exporter where circumstances change.

The procedure does not apply in the same way to organisms for food, feed or processing, which have a separate simplified procedure, nor to contained use. This is the commonest examination trap in the topic.

5.4 Risk assessment, documentation and emergencies

Risk assessment must be carried out in a scientifically sound manner, and the Party of import may require the exporter to carry it out and may charge the cost to the exporter. **Risk management** measures follow. Requirements for **handling, transport, packaging and identification** attach documentation to shipments, with the level of specificity differing between the three categories.

For **illegal and unintentional transboundary movements**, the Party must notify affected or potentially affected States, the Biosafety Clearing House and relevant international organisations, and consult on the response. This is the emergency arm of the Protocol.

5.5 The Biosafety Clearing House

Established by **Article 20(1)** as part of the Clearing House Mechanism of the Convention, to facilitate the exchange of scientific, technical, environmental and legal information on living modified organisms and to assist Parties, **with particular regard to the special needs of developing country Parties, least developed countries, small island developing States, countries with economies in transition, and countries that are centres of origin and centres of genetic diversity**. Each Party designates a national focal point.

1. **Two kinds of record**: national records mandated by Article 20, and reference records.
2. **Six official languages**: English, French, Spanish, Russian, Arabic and Chinese.
3. **Unique Identifiers Registry**, holding summary information on all registered organisms including transformation events, genetic modifications and the unique identification code.
4. **Gene Registry**, holding gene inserts and the characteristics of the modifications.
5. **Organism Registry**, holding parental, recipient and donor organisms.

The connection to draw out: **access and benefit sharing under the Convention is impossible unless you can trace where a genetic resource came from, and the Organism Registry gives you the lineage. So the biosafety instrument also serves the third objective of the Convention, the fair and equitable sharing of benefits.**

5.6 Precaution and supporting mechanisms

The Protocol expressly **reaffirms the precautionary language in Principle 15 of the Rio Declaration on Environment and Development**. It is supported by **capacity building, socio economic considerations, a compliance procedure, liability and redress**, and the **Conference of the Parties serving as the meeting of the Parties**. Liability and redress were developed in the **Nagoya Kuala Lumpur Supplementary Protocol** of 2010, in force from 2018, which operates administratively: the competent authority requires the operator to take response measures, or takes them and recovers the cost.

5.7 Benefits of being a Party

1. Enhanced **visibility and credibility** of the national biosafety system.
2. **Harmonised rules, procedures and practices** for transboundary movement.
3. Opportunities to **collaborate** with other governments, the private sector and civil society.
4. Improved **access to relevant technologies and data** and regular exchange of information and expertise.
5. Demonstration of **commitment to conservation and sustainable use**.
6. Ability to **influence the Protocol's development** through the meeting of the Parties.
7. **Eligibility for financial support from the Global Environment Facility** for developing country Parties and Parties with economies in transition.

5.8 Movement within a country, and India's compliance

The Protocol governs movement **between** countries. Movement **within** a country is governed by domestic law: in India by the Rules, 1989, the guidelines under them, institutional biosafety clearance, and, for transfers of material between institutions, a **Material Transfer Agreement**. **The material transfer agreement is the private law instrument that allocates risk domestically in the way the Protocol does internationally.**

On compliance, India's position is mixed and an answer should say so. India is a party and the Rules, 1989 with the guidelines under them substantially implement the notification, risk assessment and containment obligations, and the import of any genetically engineered organism requires appraisal committee permission. **What India has not done is enact domestic legislation giving effect to the liability and redress regime of the Nagoya Kuala Lumpur Supplementary Protocol,** and it has not finalised its labelling and import tolerance rules for genetically modified food, which are the two clearest compliance gaps.

PART 6 REGULATORY AGENCIES FOR GM CROP, FOOD AND MEDICINE, AND THEIR FUNCTIONING

6.1 How to map a product to its regulator

Begin every answer here, because the product determines the statute and the statute determines the agency.

1. A **genetically engineered plant** engages the Environment (Protection) Act, 1986 and the Rules, 1989, so the institutional, research and appraisal committees apply.
2. A **food**, including one derived from such a plant, engages the Food Safety and Standards Act, 2006 and the food authority.
3. A **medicine or device** engages the Drugs and Cosmetics Act, 1940 and the central drug regulator with the State licensing authorities.
4. A **biologic**, being a medicine made using a genetically engineered organism, engages **both** the 1940 Act and the Rules, 1989 at the same time.
5. A **stem cell product** is a drug, so it engages the 1940 Act, plus the stem cell guidelines and the ethics committee framework.
6. An **animal experiment** engages the Prevention of Cruelty to Animals Act, 1960 and the central and institutional animal committees.
7. **Human participants** engage the New Drugs and Clinical Trials Rules, 2019 and the registered ethics committees.

6.2 Agencies for genetically modified crops, and how they function

The six authorities are set out in Part 4A. Their functioning in practice has four features.

1. **Sequential dependency.** No stage can begin before the previous one concludes, and failure at a later stage can send the applicant back to an earlier one.
2. **No single window.** Unlike the European Union's single door, **India has no coordinating interface, no statutory timeline for the chain as a whole, and no obligation on one agency to defer to another.**
3. **Split ministries.** The approval body sits under the environment ministry while the research committees sit under the Department of Biotechnology, which also promotes the industry.
4. **A monitoring tier that under performs.** State and district committees exist on paper; illegal cultivation shows they do not function uniformly.

6.3 Agencies for food

The **Food Safety and Standards Authority of India** was created by the Food Safety and Standards Act, 2006 as an independent statutory regulator. Four functions matter here.

1. **Section 22** of the Act governs genetically modified and engineered foods and is the source of its jurisdiction; in practice it prohibits manufacture, distribution, sale or import of such food without approval.
2. **Import control**, including the requirement of a non genetically modified and GM free certificate for specified food crops applied from 1 January 2021, clarified as not extending to processed foods generally.
3. **Draft Genetically Modified or Engineered Foods Regulations**, consulted upon in 2021 and 2022 and not finally notified, which is why India lacks an operative labelling regime.
4. **Enforcement** through State food safety commissioners and designated officers, mirroring the Centre and State split in drug regulation.

The **Supreme Court in July 2024 directed that the provisions of the Food Safety and Standards Act be enforced in matters of import of genetically modified food**, which is the clearest judicial statement yet on the division of labour between the two regulators.

6.4 Agencies for medicines: India

The **Central Drugs Standard Control Organisation** operates under the Ministry of Health and Family Welfare, headquartered in New Delhi with **six zonal and sub zonal offices**, and is headed by the **Drugs Controller General of India**, who acts as the Central Licensing Authority for new drugs, clinical trials, imports and higher risk devices. Supporting statutory bodies are the **Drugs Technical Advisory Board** under section 5 for technical advice, the **Drugs Consultative Committee** under section 7 for Centre and State coordination, and the **Central Drugs Laboratory** under section 6 as the statutory testing laboratory. **State Licensing Authorities** license ordinary drugs and enforce through drugs inspectors, and prices are set by the **National Pharmaceutical Pricing Authority** under the Drugs (Prices Control) Order, 2013.

The context for its expansion is that **India supplies approximately 20 per cent of global generic medicines by volume to more than 200 countries.**

How the system now functions

1. **The 2019 paradigm shift.** The New Drugs and Clinical Trials Rules, 2019 replaced provisions poorly adapted to modern biologics and international submissions, strengthening oversight, introducing **defined timelines** and mandating digitised workflows.
2. **The SUGAM portal** provides real time application tracking, document submission and status transparency.
3. **The clinical trial waiver. Rule 101** empowers the Central Licensing Authority, with the approval of the Central Government, to specify countries for considering waiver of local clinical trials. By **order dated 7 August 2024** the regulator specified **six jurisdictions: the United States, the United Kingdom, Japan, Australia, Canada and the European Union. The waiver applies to five categories: orphan drugs for rare diseases; gene and cellular therapy products; new drugs for pandemic situations; new drugs for special defence purposes; and new drugs offering significant therapeutic advance over the current standard of care.** Phase IV trials remain mandatory. *Two corrections to the course slides: this was an order issued under Rule 101, not an amendment of Rule 101, and the list is six jurisdictions, not four.*
4. **Why the waiver mattered.** India had for decades required local Phase III trials even where extensive global data existed, justified by pharmacogenomic differences across a population of 1.4 billion. The cost was a **drug lag**, with development lifecycles in India of twelve to eighteen years for some products.
5. **Registration of clinical research organisations** became mandatory in 2024, closing the gap that arose because the practical conduct of trials is outsourced to them.
6. **Residual weaknesses.** Pharmacovigilance is less systematised than in the United States or European Union; the review process offers **less public transparency**, though the portal has improved this; and **State licensing authorities retain significant power** over manufacturing licences and local enforcement, creating variability that a single federal regulator does not face.

6.5 Agencies for medicines: United States and European Union

The **Food and Drug Administration** traces its authority to the **Pure Food and Drugs Act of 1906**, a response to widespread adulteration and mislabelling. It functions as a **single centralised federal authority** regulating drugs, biologics, devices and food across all fifty States, operating through **Title 21 of the Code of Federal Regulations**, which defines standards for manufacturing, labelling, clinical trials and postmarket surveillance.

The **European Medicines Agency** provides the scientific foundation, coordinating the perspectives of more than twenty seven Member States, through three routes: the **centralised procedure**, where one application grants authorisation valid in all Member States and which is mandatory for biologics and new active substances; the **decentralised procedure**, with simultaneous applications and one Reference Member State; and **mutual recognition**, where authorisation in one State is recognised by others.

The structural contrast is worth stating. A single regulator trades speed for distance from local conditions; a network preserves autonomy at the cost of complexity; **India's Centre and State split suffers from neither of those but from uneven enforcement capacity.**

6.6 How the three medicine systems differ in operation

1. **Approval timelines.** In the 2011 to 2015 period the American regulator averaged about **eighteen months** while the European and Indian regulators averaged about **twelve**. The longer American average reflects the depth of review and more extensive clinical data packages. Treat these as a dated snapshot.
2. **Volume.** Across 2011 to 2015 the three jurisdictions approved **509 new drugs**: the European agency **257**, the American agency **182** and India **70**, before the reforms from 2019 accelerated Indian throughput.
3. **Batch validation.** The American regulator accepts a **single batch** for process validation at submission; the European regulator requires **three consecutive batches**. Western markets generally require a **minimum batch size of 100,000 units**.
4. **Stability data.** The American regulator requires **three months** of accelerated and long term data at submission; the European regulator **six months** of accelerated data; India **six months**, applying **tropical Zone IV conditions of 40 degrees Celsius and 75 per cent relative humidity**.
5. **Finished product specifications.** Assay standards are reported as 90 to 100 per cent in the United States, 95 to 105 per cent in the European Union and 90 to 110 per cent in India, with disintegration testing mandatory in the latter two and case by case in the first. **A product can be lawful in one country and unlawful in another with no difference in safety, which is a pure regulatory barrier to trade.**

6.7 How the agencies assess generics and biologics

For **generics**, bioequivalence studies are the cornerstone, showing that the test product delivers the same active ingredient at the same rate and extent as the reference product. Four design requirements are common to all three jurisdictions.

1. **Fasted studies** are mandatory everywhere as the standard condition.
2. **Fed studies** are required for modified release formulations and food effect assessment.
3. **Sample retention.** The United States mandates **five year retention** of biological samples, enabling retrospective verification.
4. **Statistical criterion.** The **90 per cent confidence interval of maximum concentration and area under the curve must fall within 80 to 125 per cent** in all three regions.

For **biologics**, **the process is the product: even small manufacturing changes can alter clinical behaviour**, so they cannot be identically replicated, only made highly similar. The European Union was the first major jurisdiction to establish a biosimilar pathway, publishing its first guideline in **2005**, and its approach emphasises **clinical similarity through a totality of evidence assessment** rather than strict chemical identity. The United States pathway came with the **Biologics Price Competition and Innovation Act of 2010**, distinguishing a **biosimilar** from the more stringent **interchangeable** designation permitting substitution at pharmacist level. India relied historically on **Schedule Y** with international guidelines, approving over twenty biogeneric products, before publishing dedicated **Guidelines on Similar Biologics in 2012, revised in 2016**.

Three technical points explain why biologics need different regulatory science.

1. **Preclinical safety.** Standard rodent models often fail to predict how complex proteins behave in humans. A safe starting dose requires a **minimal anticipated biological effect level** derived from in vitro receptor occupancy and in vivo dose response in a **pharmacologically relevant species**, which for many biologics means non human primates, raising its own ethical problem.
2. **Immunogenicity.** All biopharmaceuticals can trigger **anti drug antibodies**, which may neutralise the effect, alter pharmacokinetics or cause severe reactions including anaphylaxis. Validated assays with prespecified thresholds are required, and **protein aggregation, oxidation and deamidation over shelf life amplify the risk.**
3. **The genotoxicity gap.** The standard battery of the Ames test, in vitro chromosomal aberration and in vivo micronucleus assay was designed for small molecules and is **largely irrelevant for biologics, which are too large to enter cells and do not interact directly with DNA.** Regulators accept mechanistic justifications for waiving it and substitute **adventitious agent testing and cell bank characterisation.**

A biologic in India therefore engages **two regulators at once**, the drug regulator under the 1940 Act and the recombinant DNA authorities under the Rules, 1989. **A strong criticism follows: India regulates its most complex medicines mainly through non statutory guidelines.**

6.8 Agencies for stem cells, animals and human participants

1. **Stem cells.** The drug regulator, because stem cells are drugs; the **Cell Biology Based Therapeutic Drug Evaluation Committee** for cell therapy trials; the institutional ethics committee; and, until the March 2024 dissolution, the institutional and national stem cell committees. Import needs a licence, and international collaboration needs the **Health Ministry's Screening Committee.**
2. **Animals.** The central animal committee under section 15 of the 1960 Act, with **Institutional Animal Ethics Committees** below it, each including a nominee of the central committee.
3. **Human participants.** Registered ethics committees under the 2019 Rules, applying the ICMR National Ethical Guidelines of 2017.
4. **Gene editing of human cells** additionally requires the **Review Committee on Genetic Manipulation**, which is how the biosafety machinery reaches into clinical research.

6.9 Post approval: where the systems really diverge

Approval is the starting gun, not the finish line. **The most consequential divergences show up not in pre approval dossiers but in the maturity and data infrastructure of post market pharmacovigilance.**

1. **United States.** A mature systematised ecosystem: **FAERS** for spontaneous reporting, **Sentinel** for active surveillance in real world data, and **Risk Evaluation and Mitigation Strategies** imposing specific safety measures as conditions of approval for high risk products.
2. **European Union.** Safety monitoring is a **lifecycle obligation: Periodic Safety Update Reports, Risk Management Plans** as a mandatory approval condition, and **post authorisation safety studies**, with **EudraVigilance** enabling cross border signal sharing.
3. **India.** The **Pharmacovigilance Programme of India**, launched in 2010, coordinates adverse event reporting between State and central authorities, but faces **underreporting, healthcare infrastructure gaps and complex distributed supply chains.**

Two further gaps. **India lacks a standalone mandatory recall law with defined obligations, timelines and escalating sanctions,** which is a structural weakness in a fragmented supply chain of hundreds of manufacturers; a dedicated recall statute would define communication protocols, link removal to adverse event monitoring and establish measurable accountability. And the American regulator has committed to expanding **unannounced international inspections**, which matters because a substantial proportion of the drugs and active ingredients consumed there are manufactured in India and China.

On quality, **the most consequential evolution is the shift from checklist compliance to quality systems thinking: not did you submit the document, but does your organisation have the culture, capability and infrastructure to produce safe medicines consistently.** India's revised **Schedule M** requirements of 2023 introduce stronger quality management expectations, including premises and equipment standards, critical process validation, mandatory self inspection and documentation discipline aligned with international practice.

6.10 Harmonisation and the direction of travel

The **International Council for Harmonisation** drives convergence, and three of its outputs matter most: **E6 on good clinical practice**, now in its third revision introducing risk proportionate oversight, remote monitoring and decentralised trial models; **Q8 to Q10 on quality by design**, moving from fixed specifications to design spaces and continuous verification; and the **Common Technical Document** in five modules, being **regional, summaries, quality, nonclinical and clinical**, with the transition to the electronic format enabling faster review and automated validation. The Council has more than thirty member and observer regions, and **India's drug regulator participates as an observer.**

The emerging paradigm is **regulatory reliance**, meaning that authorities recognise and build on each other's scientific work rather than repeating it. The aspirational endpoint is a **regulatory passport**, a single rigorous review granting access across jurisdictions. **Reliance requires more than goodwill: it demands compatible data systems, transparent review processes and bilateral agreements defining the scope and limits of recognition.** The realistic destination is **pragmatic convergence, not uniformity**: aligned scientific expectations, locally calibrated public health policy, sustained lifecycle pharmacovigilance and inspection ready manufacturing oversight.

6.11 The role of the legal professional

1. **Statutory interpretation.** Interpreting the interplay between the Environment (Protection) Act, 1986 and the Food Safety and Standards Act, 2006, identifying which prevails for genetically modified food crops, and advising on compliance satisfying both regimes simultaneously.
2. **Administrative advocacy.** Navigating the six authority web, filing applications, managing inter agency coordination, representing clients before the appraisal committee, and **challenging unreasonable delay through writ jurisdiction.**
3. **Trade law counselling.** Advising on compliance with the sanitary and phytosanitary agreement, on asynchronous approval risk and on treaty obligations.
4. **Proactive regulatory design.** Moving beyond litigation into policy design, and contributing to the national policy framework mandated by the 2024 Supreme Court directive.

APPENDIX A CORRECTIONS TO THE COURSE MATERIAL

The slides and lecture notes contain errors and dated statements. Reproducing them in an answer would cost marks.

1. **Declaration of Helsinki.** The slides say the last amendment was in 2000. **The current version is the revision of 19 October 2024, the eighth revision.**
2. **CRISPR mushroom.** The deck names the developer as Yiping Qi. **The developer was Yinong Yang, a plant pathologist at Pennsylvania State University;** the APHIS letter is dated 13 April 2016.
3. **The 2017 Supreme Court mandate on genetically modified food.** The deck describes a judicial direction. **What the record shows is an affidavit filed by the food authority in the Supreme Court in May 2017** setting out the division of responsibility under section 22 of the Food Safety and Standards Act.
4. **Rule 101 clinical trial waiver.** The deck says the 2024 amendment to Rule 101 recognised four authorities. **It was an order dated 7 August 2024 issued under Rule 101, not an amendment, and it specified six jurisdictions.**
5. **Dickey Wicker Amendment.** The lecture dates it to 1995 in one place and 1996 in another. **The correct year is 1996.**

6. **Animal Welfare Act.** The lecture calls it the Act of 1985. **It was enacted in 1966; 1985 is the year of the amendment** introducing the listed requirements.
7. **CPCSEA.** The lecture uses the old name and ministry. **The body is now the Committee for Control and Supervision of Experiments on Animals, under the Department of Animal Husbandry and Dairying since April 2019.**
8. **Stem cell oversight committees.** The lecture treats the institutional and national stem cell committees as mandatory. **That two tier structure was dissolved in March 2024,** and review now sits with institutional ethics committees.
9. **Biologics Price Competition and Innovation Act.** Dates in the material vary between 2009 and 2010. **The Act was enacted in 2010.**
10. **Number of committees under the Rules, 1989.** Some notes say four. **There are six.**
11. **Nuremberg charges.** Some notes name three charges while claiming four. Say that the charges included conspiracy, war crimes, crimes against humanity and membership of a criminal organisation, and do not invent a fourth head.

APPENDIX B REVISION SHEET

B.1 Numbers that carry marks

Nuremberg: **10 points**, drafted 9 August 1947, verdict 20 August 1947, **6 expert points expanded to 10**, 23 accused. Helsinki: **1964, 32 principles** in the 2000 version, current revision **19 October 2024**. Belmont: **3 principles, 3 applications**; Tuskegee **399 men, 28 deaths, 100 disabled, 40 wives, 19 congenital cases**, exposed **1972**. CIOMS: founded **1949** by WHO and UNESCO, guidelines **1982**. Common Rule: **1991**, Subparts **B, C and D**. ICMR: policy statement **February 1980**; Delhi cervical study **1,158 women, 71 malignancies, 9 invasive, 62 treated late**; controversy public **1997**; guidelines **2000** with **12 principles**; current guidelines **2017**. Ethics committee: **5 to 15 members**. IACUC: **3 criteria**. Rules 1989: **6 authorities, 5 activities**. Bt cotton: approved **2002, 95 per cent** area by 2014, pesticide use down **50 per cent**, yield up **24 per cent**. Bt brinjal: appraisal committee **14 October 2009**, moratorium **9 February 2010**, consultations in **7 cities**. GM mustard: appraisal **18 October 2022**, ministry **25 October 2022**, judgment **23 July 2024, 5 agreed directions**, consultation within **4 months**. Technical Expert Committee: constituted **2012**, report **2013, 10 year moratorium** recommended. Cartagena: adopted **29 January 2000**, in force **11 September 2003, 3 categories, 2 exemptions, 4 AIA steps, 3 registries, 6 languages**. European labelling threshold: **0.9 per cent**. World Trade Organization reports: **29 September 2006**, adopted **21 November 2006**, moratorium **June 1999 to August 2003, 24 of 27** products. Bioequivalence: **80 to 125 per cent** at **90 per cent** confidence. Clinical trial waiver: **6 countries, 5 categories**, order of **7 August 2024**. Stem cells: **14 day limit, 7 prohibited areas**, follow up **at least 2 years**.

B.2 Twelve traps

1. Saying Helsinki was last amended in **2000**. It was revised in October 2024.
2. Giving Belmont **four** principles. It has three, each with a matching application.
3. Saying the Rules, 1989 create **four** committees. There are six.
4. Saying the advance informed agreement procedure applies to **all** living modified organisms. It applies to the first intentional movement for release into the environment.
5. Treating **products derived from** living modified organisms as covered by the Cartagena Protocol. They are exempt if they no longer contain a living organism.
6. Saying Bt brinjal was **banned**. It is under an indefinite moratorium, expressly not a ban.
7. Saying the Supreme Court in 2024 **banned or approved** GM mustard. It split and referred the matter to a larger Bench.
8. Saying the Supreme Court **mandated** FSSAI jurisdiction in 2017. That was an affidavit, not an order.
9. Saying the **commercial use of stem cells** as therapy is permitted in India. It is prohibited, and no administration is permissible outside a clinical trial.

10. Confusing **section 3(b)**, morality, with **section 3(d)**, new forms of known substances.
11. Saying India regulates genetically modified crops under an **agricultural** statute. It uses an environmental one.
12. Saying a **central GEAC clearance** is sufficient for cultivation. A State may still block it, and whether it lawfully may is undecided.

B.3 Six questions these topics are designed to produce

1. Trace the development of international ethical codes from Nuremberg to the present and assess their influence on Indian law.
2. Discuss the ethical theories in animal research and evaluate the adequacy of the Indian regulatory structure.
3. What is bioethics? Examine the ethical and legal issues in stem cell research with reference to the Indian guidelines.
4. Critically examine the regulatory framework for genetically modified crops in India and compare it with the United States and the European Union.
5. Discuss the legal controversies surrounding Bt cotton and Bt brinjal, with particular reference to State bans and the question of liability.
6. Explain the salient features of the Cartagena Protocol on Biosafety and assess India's compliance with it.